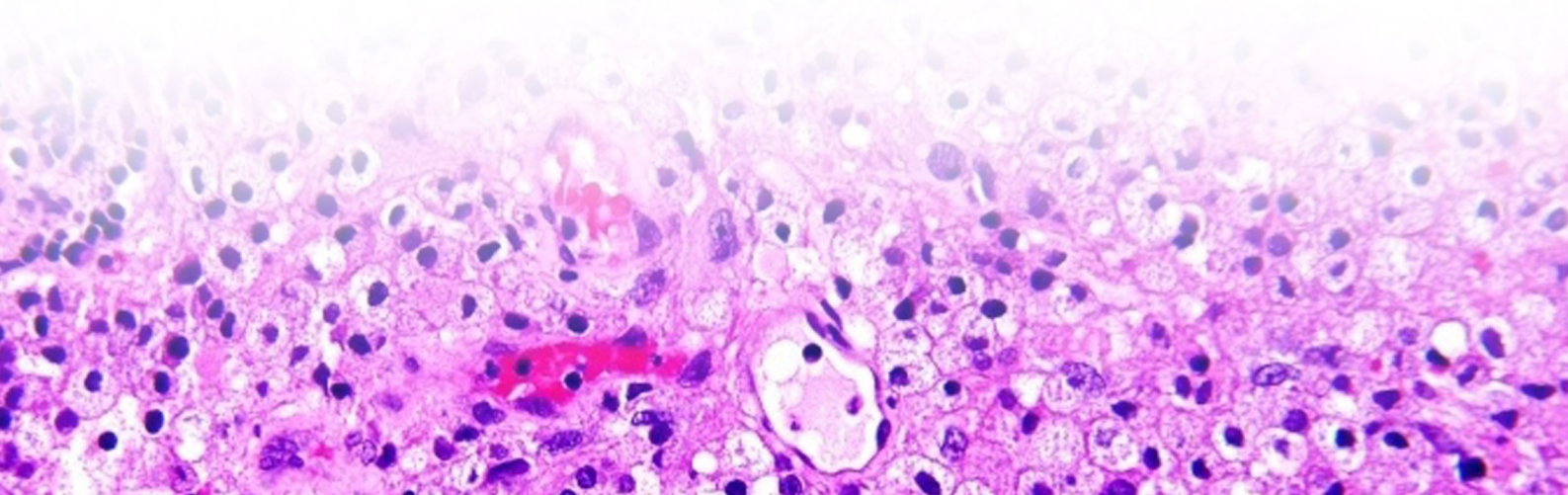
Student Notes

PATHOLOGY

1st Edition

Unit 6

Neoplasia



Preface

Agam is a group of budding medicos, who are currently doing their under graduation in various Medical Colleges across Tamil Nadu and Pondicherry. The group was initiated on 18th November 2017, in the vision of uniting medicos for various social and professional causes.

We feel delighted to present you Agam Pathology notes prepared by Agam Divide and Rule 2020 Team to guide our fellow medicos to prepare for university examinations.

This is a reference work of 2017 batch medical students from various colleges. The team took effort to refer many books and make them into simple notes. We are not the authors of the following work. The images used in the documents are not copyrighted by us and is obtained from various sources.

Dear readers, we request you to use this material as a reference note, or revision note, or recall notes. Please do not learn the topics for the 1st time from this material, as this contain just the required points, for revision.

Acknowledgement

On behalf of the team, Agam would like to thank all the doctors who taught us Pathology. Agam would like to whole heartedly appreciate and thank everyone who contributed towards the making of this material. A special thanks to Vignesh M, who took the responsibility of leading the team. The following are the name list of the team who worked together, to bring out the material in good form.

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NEOPLASIA

ESSAY:

1. Invasion and metastasis of cancer.
2. Difference between benign and malignant tumor,
3. Define neoplasia. Molecular basis of cancer. Add a note on oncogenes
4. Chemical carcinogenesis
5. Physical carcinogenesis
6. Biological carcinogenesis.
7. Pathways of spread of tumor

SHORT NOTES:

1. Differentiation and Anaplasia.
2. Etiological agents of cancer.
3. Metaplasia
4. Precancerous lesions
5. p53
6. Warburg effect
7. Growth factors
8. Retinoblastoma
9. Tumor markers
10. Exfoliative cytology
11. Oncofetal protein
12. Lab diagnosis of Cancer
13. FNAC
14. Autonomic tissue processor
15. Paraneoplastic syndrome
16. Grading and Staging of tumour
17. Carcinoma in situ

SHORT ANSWERS:

1. Immune evasion by cancer
2. Dysplasia
3. Hamartoma
4. Carcinoma in situ
5. Metastasis

UPDATES

ESSAY

1. INVASION–METASTATIC CASCADE:

Invasion and metastasis are characteristic of malignant tumors.

DEFINITION:

- Invasion–metastatic cascade constitutes the entire sequence of events from the beginning of invasion to the development of metastasis.
- Invasion: Second most important criteria for malignancy.

PHASES:

Invasion–metastatic cascade is a complex multistep process. It can be divided into two main phases, namely:

- Invasion of the extracellular matrix (ECM)
- Metastasis (vascular dissemination and homing of tumor cells).

(A) INVASION OF EXTRACELLULAR MATRIX:

Invasion of the ECM is an active process and consists of four steps:

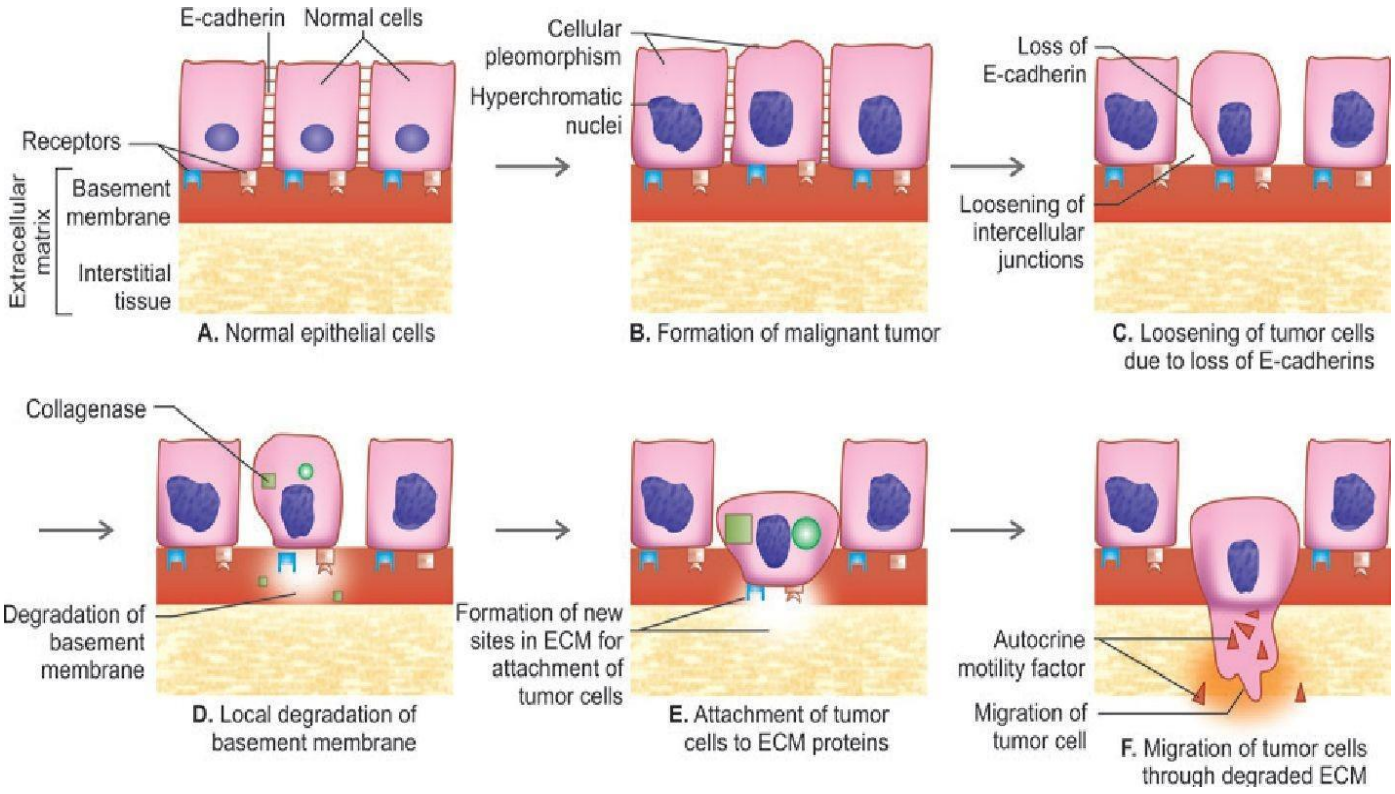
- **Loosening of tumor cells:**
  - ✓ Normal cells are attached to each other by adhesion molecules namely E-cadherins.

- **Reduced/loss of E-cadherin function:**
  - ✓ It is observed in most epithelial cancer (e.g. adenocarcinomas of the colon and breast), there is loosening of tumor cells.
  - ✓ The separated cells get detached from the primary cancer.

- **Local degradation/proteolysis of basement membrane and interstitial connective tissue:**
  Extracellular matrix is of two types, namely:
    - ✓ Basement membrane
    - ✓ Interstitial connective tissue.
  - **Secretion of degrading enzymes:** Malignant tumor cells and stromal cells (e.g. fibroblasts and inflammatory cells) in the cancers secrete / induce many proteolytic enzymes that degrade ECM. These enzymes include:
    - ✓ Matrix metalloproteinases (MMPs),
    - ✓ Cathepsin
    - ✓ Urokinase - plasminogen activator (u-PA).
  - **Local degradation of basement membrane and interstitial connective tissue:** This is achieved by proteolytic enzymes.

- **Changes in attachment/adhesion of tumor cells to ECM proteins:** Normal epithelial cells have receptors (e.g. integrin) for basement membrane components (e.g. laminin and collagen) and are located at their basal surface.
  - **Generation of new sites:** Local degradation of basement membrane generates new and strange sites in the basement membrane.
  - **Adhesion of tumor cells to ECM:** The receptors on tumor cells attach to the new sites in the basement membrane.
  - **Stimulation of tumor cell migration:** It follows attachment/adhesion of tumor cells to ECM proteins.

- **Locomotion/migration of tumor cells through degraded ECM:** It is a multistep process.
  - **Locomotion/migration drives the tumor cells forward** through the degraded basement membranes and zones of proteolysis in the interstitial connective tissue matrix.
  - **Locomotion involves many receptors and signaling proteins**. The locomotion is potentiated by tumor cell-derived cytokines, such as autocrine motility factors (AMF) and other molecules.
  - **Migration through interstitial tissue:** The tumor cells invade and traverse through the surrounding interstitial connective tissue and ultimately reach nearby blood and lymphatic vessels. Cells gain access to the circulation by penetrating the basement membrane of vessels.



(B) METASTASIS:

Metastasis is the process of deposition of tumor deposits away from primary tumor. Following the invasion of surrounding interstitial tissue, malignant cells may spread to distant sites by metastasis. Metastasis is multistep process by which tumor produces a secondary growth at a distant site or location.

- **Penetration of vascular or lymphatic channels** (intravasation into the lumen of vessels):
  - Malignant cells penetrate the basement membrane of blood vessels or lymphatic channels.
- **Invasion of the circulation and formation of tumor emboli:**
  - In the circulation, tumor cells are susceptible to destruction by several mechanisms, such as,
    - ✓ Mechanical shear stress,
    - ✓ Apoptosis stimulated by loss of adhesion (termed *anoikis*)
    - ✓ Innate and adaptive immune defenses.
  - Survived tumor cells within the circulation, may clump with platelets to form platelet tumor aggregates. This may enhance tumor cell survival and implantability.
  - Tumor cells may also bind and activate coagulation factors and form emboli.
- **Transit through the circulation.**
- **Arrest within circulating blood or lymph:**
  - It occurs at distant location away from primary tumor.
  - At the site of arrest tumor cells adhere to endothelial cells.
- **Exit from the circulation into a new tissue site:**
  - Location at which circulating tumor cells leave the capillaries to form secondary deposits depends on the anatomic location and vascular drainage of the primary tumor and the tropism of particular tumors for specific tissues.
  - Exit occurs through the basement membrane of lymphatics or blood vessel.
  - The site at which circulating tumor cells leave the vessel or lymphatics must repeat the same events involved in invasion but in a reverse order.
- **Formation of micro metastases:** Tumor cells lodge at a distant new site to form micro metastases. Examples for favored sites of metastasis:
  - Prostatic carcinoma to the bone.
  - Bronchogenic carcinomas to the adrenals and to the brain.
  - Neuroblastomas to the liver and bones.
- **Angiogenesis.**
- **Local growth of micro metastases into macroscopic tumor.**

2) DIFFERENCE BETWEEN BENIGN AND MALIGNANT TUMOUR:

BENIGN TUMOUR	MALIGNANT TUMOUR
A tumor is said to be benign when it is localized, will not spread to the other sites.	Malignant tumors can invade and destroy adjacent structures and spread to distant sites (metastasize) to cause death.
Benign tumors are designated by attaching the suffix –oma to the name of the cell type from which the tumor originates. Tumor of mesenchymal cells follows this rule.	The nomenclature of malignant tumors essentially follows the same schema used for benign tumor. Malignant tumors arising in the solid mesenchymal tissues are usually called sarcomas and those from epithelial cells are called carcinomas.
Example: fibroma, chondroma	Example: Fibrosarcoma, chondrosarcoma

DIFFERENTIATION

BENIGN TUMOUR	MALIGNANT TUMOUR
Is always a well-differentiated tumour	Wide range of differentiation of parenchymal cells
Well-differentiated: The neoplastic cell closely resembles the normal cell of origin. It may be not possible to recognize it as a tumor by microscopic examination of individual cells (e.g. lipoma). Only the growth of these cells into discrete lobules discloses the neoplastic nature of the lesion For example: • Lipoma • Leiomyoma of the uterus.	**Well-differentiated tumors / low grade cancer:** Well-differentiated adenocarcinoma of the colon. It shows cancerous glands that are irregular in shape and size invading the muscular layer of the colon Squamous cell carcinomas of skin. The tumor consists of cells which are similar to normal squamous epithelial cells, with intercellular bridges and keratin pearls.
	Poorly differentiated tumors / high grade cancer: They consist of cells that have little resemblance to the cell of origin.
	Moderately differentiated / intermediate between high and low grade cancer: These tumors show differentiation in between the well and poorly differentiated tumors.

FEATURES:

PARAMETER	BENIGN TUMOUR	MALIGNANT TUMOUR
GROWTH	Slow growing	Rapid growing
LOCAL INVASION	No invasion, infiltration of the surrounding normal tissues	Locally invasive, infiltrate surrounding normal tissues
METASTASIS	Absent	Frequent
PROGNOSIS	Good	Poor

BASED ON HISTOLOGIC FEATURES:

PARAMETER	BENIGN TUMOUR	MALIGNANT TUMOUR
GROWTH	Organized growth	Disorganized growth [loss of potency]
NUCLEAR MORPHOLOGY	Uniform nuclei, normal	Nuclear pleomorphism, hyperchromasia and irregular outline
N:C RATIO	Low	High
MITOTIC ACTIVITY	Rare, normal mitotic figures	High and abnormal mitotic figures
INVASION	Lack of invasion	Invasion present
METASTATIC SPREAD	No metastatic potential	It is the **hallmark** of malignancy
NUCLEOLI	Absent	Usual and prominent
TUMOUR GIANT CELLS	Not seen	May be seen, Nuclear atypia
CHROMOSOMAL ABNORMALITY	Not found	Usually seen
POLARITY	Maintained	Lost

BASED ON GROSS APPEARANCE:

PARAMETER	BENIGN TUMOUR	MALIGNANT TUMOUR
BORDER/CAPSULE	Mostly circumscribed and encapsulated	Poorly defined
AREAS OF NECROSIS	Rare	Common

3. NEOPLASIA:

"Neoplasia" means new growth. It is defined as a disorder of cell growth that is triggered by a series of acquired mutations affecting a single cell and its clonal progeny.

MOLECULAR BASIS OF CANCER

- Non-lethal damage lies at the heart of carcinogenesis
- Carcinogenesis results from accumulation of complementary mutation in a stepwise manner
- Tumour is formed by clonal expansion of single precursor cells that are genetically damaged
- Four class of regulatory gene
    - Growth promoting oncogene
    - Growth inhibiting tumour suppressor gene
    - Genes regulate apoptosis
    - Genes involved DNA repair

CELLULAR AND MOLECULAR HALLMARKS OF CANCER

- Self-sufficiency in growth signal
- Insensitivity to growth inhibitory signal
- Altered cellular metabolism
- Evasion of apoptosis
- Ability to evade host immune response
- Ability to invade and metastasize
- Sustained Angiogenesis
- Limitless replicative potential

ONCOGENES:

- Have the ability to promote cell growth in the absence of normal growth promoting signal
- Encode a protein called oncoprotein
- Proto-oncogene → Mutation → Oncogene
- Oncogenes can be one of the genes controlling the following functions:
    1. Growth factors
    2. Growth factor receptors
    3. Proteins involved in signal transduction
    4. Nuclear regulatory proteins
    5. Cell cycle regulators

1. GROWTH FACTOR

PROTONCOGENE	MODE OF ACTIVATION	TUMOUR
PDGFB	Overexpression	Astrocytoma
HST	Overexpression	Bladder, Breast cancer
TGFA	Overexpression	Astrocytoma
HGF	Overexpression	Hepatocellular carcinoma

2. GROWTH FACTOR RECEPTOR

PROTONCOGENE	MODE OF ACTIVATION	TUMOUR
ERBB1 ERBB2	Mutation Amplification	Adenocarcinoma of lung Breast carcinoma
ALK	Translocation	Adenocarcinoma of lung
RET	Point mutation	Multiple endocrine neoplasia MEN2 syndrome Medullary carcinoma of thyroid
FLT3	Point mutation	Leukemia

3. SIGNAL TRANSDUCTION PROTEIN

PROTO ONCOGENE	MODE OF ACTIVATION	TUMOUR
N- RAS K- RAS H- RAS	Translocation Point mutation	Melanoma Colon cancer Bladder cancer
BRAF	Translocation Point mutation	Melanoma leukemia
ABL	Translocation Point mutation	Chronic myeloid leukemia Acute myeloid leukemia
JAK2	Translocation	Acute lymphoblastic leukemia

4. NUCLEAR REGULATORY PROTEINS

PROTONCOGENE	MODE OF ACTIVATION	TUMOUR
C- MYC	Translocation	Burkitt lymphoma
N- MYC	Amplification	Neuroblastoma
L- MYC		Small cell lung cancer

ABL
Chromosome 9

+

BCR
Chromosome 22

Fusion of BCR-ABL

↓

Activation of ABL

↓

Activation of tyrosine kinase

↓

Activation of Growth Factor Signaling

↓

Chronic Myelogenous Leukemia

MYC
Chromosome 8

+

Immunoglobulin
Chromosome 14

Fusion of Ig-MYC

↓

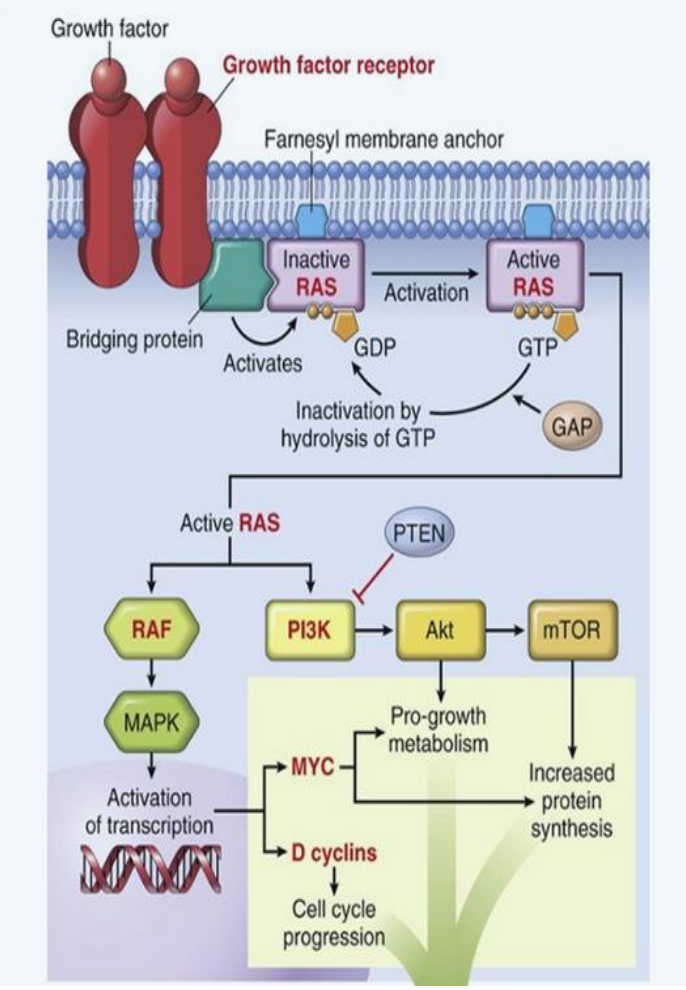
Increased MYC protein

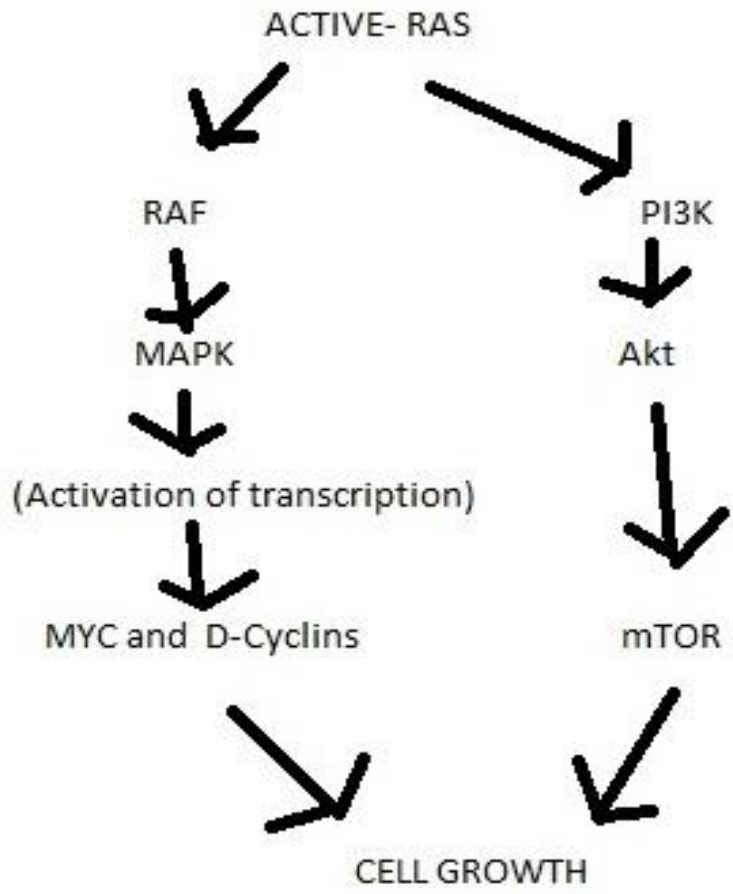
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Increased expression of pro-growth genes

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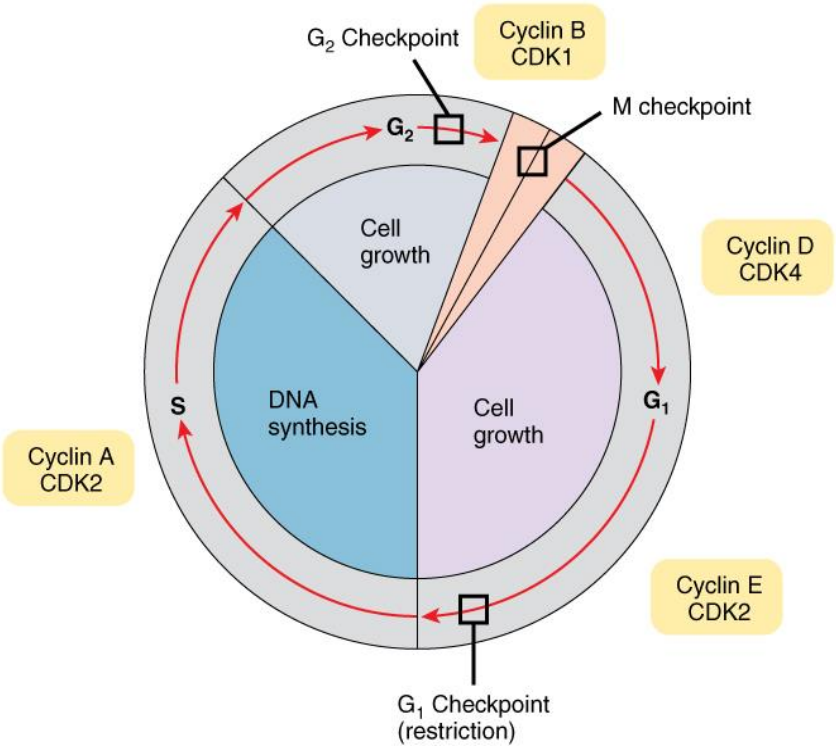
Burkitt Lymphoma





5. CELL CYCLE REGULATORS

- Cyclin D/ CDK- 4,6
- Cyclin E/ CDK- 2
- Cyclin A/CDK- 2
- Cyclin B/ CDK- 1



PROTO ONCOGENE	MODE OF ACTIVATION	TUMOUR
Cyclin D	Translocation Amplification	Multiple myeloma Mantle cell myeloma
CDK4	Amplification	Glioblastoma

CANCER ASSOCIATED MUTATION THAT AFFECT G1/ S CHECKPOINT

Gain of function mutation in cyclin D	Loss of function mutation in RB, TP53

4. CHEMICAL CARCINOGENESIS:

The induction of cancer by chemical carcinogen.
Factors: dose and mode of administration of carcinogenic chemical, individual susceptibility and other predisposing factors.

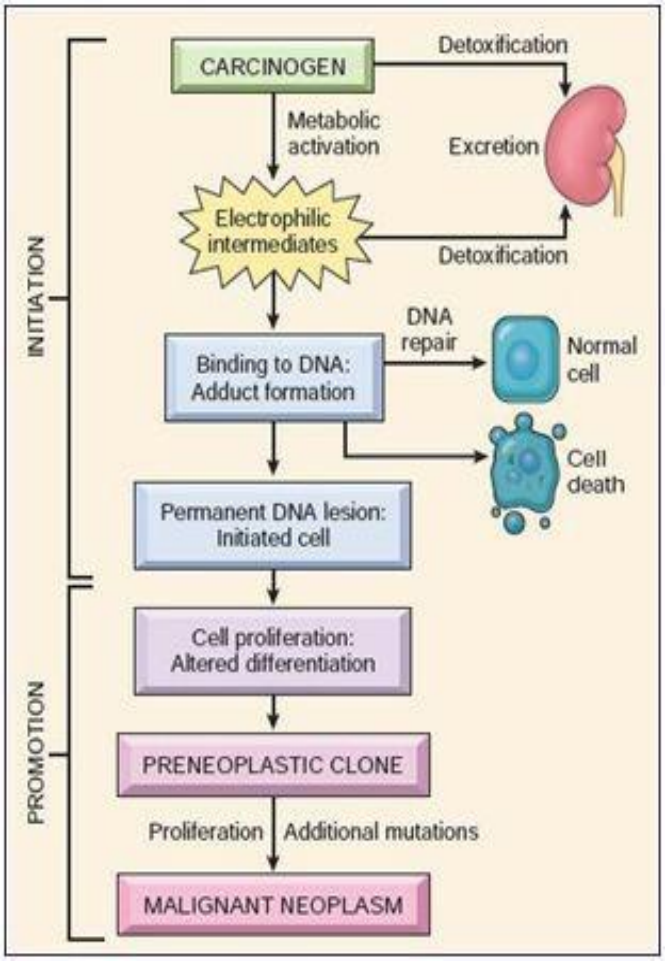
STEPS INVOLVED:

INITIATION: First stage induced by initiator chemical carcinogens, sudden, irreversible

PROMOTION: Promoters are substances like phorbol esters, phenols, hormones, artificial sweeteners, and drugs etc.., not a sudden change, not mutagenic, reversible.
Initiation alone is not sufficient for promotion

PROGRESSION: The mutated and proliferated cells show phenotypic features of malignancy, acquires additional mutations.

Initiation: causes mutagenesis.
Some carcinogens interact preferentially with particular DNA sequences or bases.
E.g.: Aflatoxin b1 > TP53 mutation in hepatocellular carcinoma.
Promotion: enhances the proliferation of cells (Mitogenic).
E.g.: unopposed estrogen stimulation of the endometrium and breast, and chronic inflammatory processes



CHEMICAL CARCINOGENS: 2 TYPES

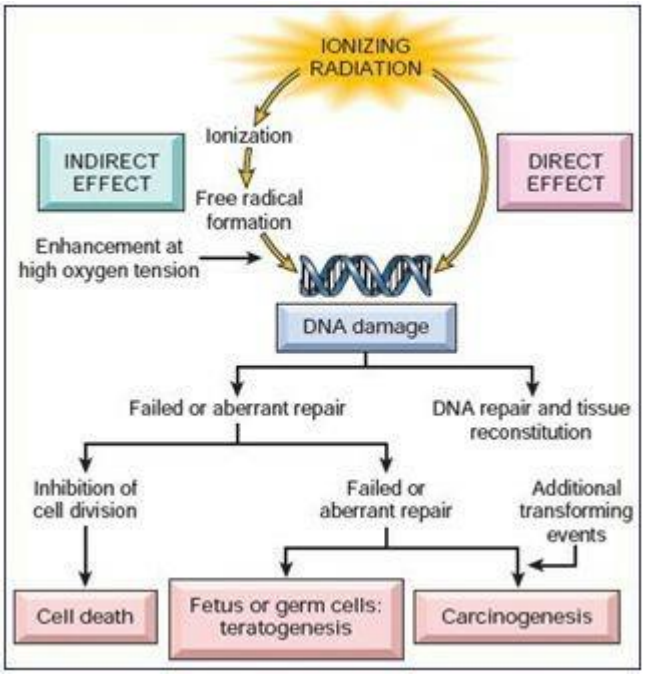
- Direct Acting
  - Do not require metabolic activation
  - Eg: β – Propiolactone, Anticancer drugs (cyclophosphamide, nitrosoureas)
- **Indirectly Acting:**
  - Require metabolic activation.
  - Mostly by highly polymorphic CYP-450 dependent mono oxygenases. Explains the susceptibility variation among people to the same carcinogen.
  - E.g.: Benzopyrene, benzidine, Aflatoxin B1, Griseofulvin.

5. PHYSICAL CARCINOGENESIS:

NON-RADIATION INDUCED:

- Mechanical injury to the tissues such as from stones in the gallbladder, urinary tract, and healed scars can be a cause of increased risk of carcinoma in these tissues
- Asbestos has been associated with some tumours of the lung (e.g.: malignant mesothelioma of the pleura)

RADIATION INDUCED:



- Radiation is a well-known carcinogen.
- Most frequent radiation-induced cancers- Myeloid leukemia.
- Others- cancers of thyroid, breast and lung.
- Lymphoid tissue: Most sensitive to radiation.
- They are more common on parts of the body regularly exposed to sunlight and ultraviolet light (UVL).
- Bone: Least sensitive to radiation.
- **UV rays cause skin cancer:**
  - Squamous cell carcinoma
  - Basal cell carcinoma
  - Malignant melanoma.
- **Latency:**
  - Extremely long latent period is common and it has a cumulative effect.
  - Radiation has additive or synergistic effects with other potential carcinogenic agents.

TYPES OF RADIATION: They are divided into two types, namely

- Ultraviolet (UV) rays of sunlight and
- Ionizing electromagnetic and particulate radiation.

RISK FACTORS: The amount of damage incurred depends on:

- Type of UV rays
- Intensity of exposure.
- Protective mantle of melanin
  - Melanin absorbs UV radiation and has a protective effect.
  - Skin cancers are more common in fair-skinned people and those living in geographic location receiving a greater amount of sunlight.

PATHOGENESIS:

- UV radiation → formation of pyrimidine dimers in DNA → carcinogenicity.
- DNA damage is repaired by the nucleotide excision repair pathway.
- With excessive sun exposure, the DNA damage exceeds the capacity of the nucleotide excision repair pathway and genomic injury becomes mutagenic and carcinogenic.
- **Xeroderma pigmentosum:**
  - It is a rare hereditary autosomal recessive disorder
  - It is characterized by congenital deficiency of nucleotide excision repair DNA.
  - These individuals develop skin cancers (basal cell carcinoma, squamous cell carcinoma and melanoma) due to impairment in the excision of UV-damaged DNA.
- **Ionizing Radiation:**
  - Eg:
    - ✓ Electromagnetic (X-rays, γ-rays) and
    - ✓ Particulate (α-particles, β-particles, protons, neutrons) radiations.
  - Damages DNA and causes genetic damage by:
    - ✓ Chromosomal breakage
    - ✓ Translocations
    - ✓ Point mutations

CANCERS PRODUCED

- **Medical or occupational exposure:** e.g. leukemia and skin cancers
- **Nuclear plant accidents:** Risk of lung cancers.
- **Atomic bomb explosion:**
  - Survivors atomic bomb explosion (dropped on Hiroshima and Nagasaki) → increased incidence of leukemia → mainly acute and chronic myelogenous leukemia after about 7 years.
  - Subsequently increased mortality due to solid tumors (e.g. breast, colon, thyroid and lung).
- **Therapeutic Radiation:**
  - Papillary carcinoma of the thyroid follows irradiation of head and neck and
  - Angiosarcoma of liver due to radioactive thorium dioxide used to visualize the arterial tree.
  - Mechanism: Hydroxyl free radical injury to DNA.
  - Tissues which are relatively resistant to radiation-induced neoplasia: Skin, bone and the gastrointestinal tract.

6. BIOLOGICAL CARCINOGENESIS (Viruses > bacteria > parasites)

VIRAL CARCINOGENESIS:

ONCOGENIC RNA VIRUSES

HUMAN T-CELL LEUKEMIA VIRUS TYPE 1 (HTLV-1)

- It is a retrovirus.
- Major target for neoplastic transformation: CD4+ T lymphocytes.
- Tumor caused: Adult T-cell leukemia/lymphoma (ATLL)
- The cancer develops after a long latent period (20–50 years).
- Mode of infection:
  - Sexual intercourse
  - Blood products
  - Breast feeding.

MECHANISM OF ONCOGENESIS:

- It is a multistep process - HTLV-1 infects CD4+ T-cells.
- HTLV-1 does not contain oncogene and its genes cannot integrate into the host genome.
- HTLV-1 genome contains gag, pol, env and a unique gene – tax
- TAX gene encodes for TAX protein which
  - Interacts with PI3K and stimulates AKT: Increased pro-growth signaling and cell survival.
  - Interferes with DNA-repair functions: Increased genomic instability
  - Is required for viral replication and cellular transformation.
  - Activates other genes involved in T-cell proliferation and differentiation.
  - These include genes that code for:
    - ✓ Interleukin (IL)-2 and its receptor (IL-2R)
    - ✓ IL-15 and its receptor IL-15R
    - ✓ Granulocyte macrophage colony-stimulating factor (GM-CSF).
      - Inactivates: p53 and other genes controlling cell cycle (e.g. CDKN2A/p16 gene).
- Secretion of cytokines and autocrine stimulation of CD4+ T-cells → proliferation of nonmalignant polyclonal cells.
- Tax protein also stimulates secretion of GM-CSF by CD4+ T-cells → stimulates nearby macrophages to produce T-cell mitogens → polyclonal proliferation of CD++ T-cells.
- TAX inactivates p53 and other genes controlling cell cycle → increased risk of developing mutations and genomic instability in proliferating CD4+ T-cells.
- Accumulation of mutations and chromosomal abnormalities → monoclonal neoplastic proliferation of CD4+ T-cell.

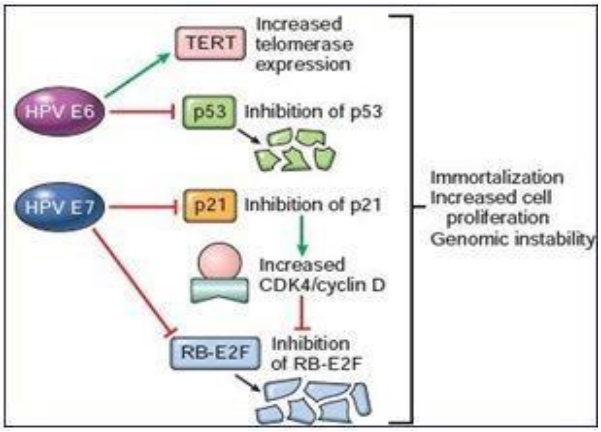
ONCOGENIC DNA VIRUSES:

HUMAN PAPILLOMAVIRUS (HPV)

- Cell infected: Human papillomaviruses (HPV) infects only the immature squamous cells but its replication occurs in the maturing, non-proliferating squamous cells.
- Their full productive life cycle occurs only in squamous cells.
- The physical state of the virus differs in different lesions.
- Types of HPV and associated lesions: More than 70 genetically different types of HPV have been identified. They are divided into
  - Low risk HPVs (HPV-1, 2, 4, and 7) cause benign squamous papilloma (warts).
  - High-risk HPVs (types 16 and 18) – cause squamous cell carcinomas of
    - ✓ Cervix,
    - ✓ Anogenital region, and
    - ✓ Head and neck.

MODE OF ACTION

- **Episomal form:** In benign lesions such as benign warts, condylomata and most precancerous lesions; the HPV genome is present as nonintegrated, free (episomal) viral DNA.
- **Integration:**
  - Intergration of HPV genome with host genome is essential for malignant transformation.
  - Integration results in overexpression of the two viral genes E6 and E7.
  - Protein products of E6 and E7 are important for the oncogenic effects of HPV.
- **Actions of E7 protein:**
  - **Inactivation of tumor suppressor RB gene:** E7 protein binds to the hypo phosphorylated (active) form of RB protein → releases its inhibitory effect on cell cycle progression
  - **Inactivation of inhibitors of cell cycle:** Inactivation of CDKIs (CDKN1A/p21 and CDNK1B/ p27) → activates cell cycle.
  - **Activation of cyclins (activators of cell cycle):** These include cyclins E and A → facilitates G2/M transition → activation of cell cycle.



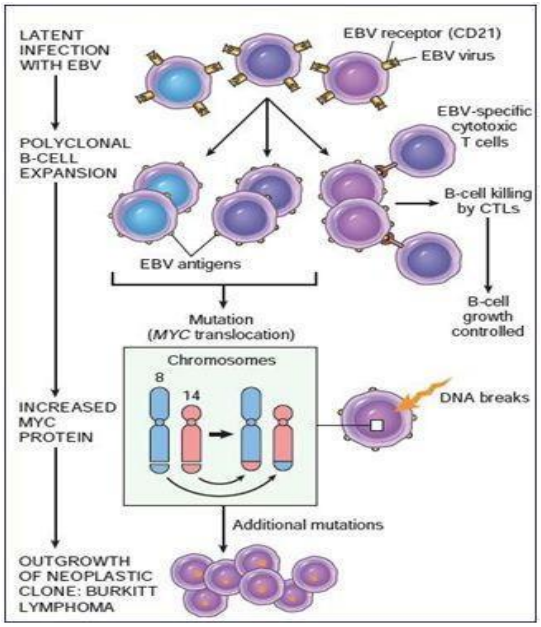
- **Actions of E6 protein:** The E6 protein complements the effects of E7.
  - **Inactivation of tumor suppressor p53 gene:** E6 binds and degrades p53 → degrades BAX (a proapoptotic factor) → prevents apoptosis.
  - **Activation of telomerase:** E6 stimulates the expression of TERT (the catalytic subunit of telomerase → prevents replicative senescence and cell proliferation continues.
- **Combined action of E6 and E7:** They induce centrosome duplication and genomic instability.

EPSTEIN-BARR VIRUS (EBV)

- EBV is a human herpesvirus, which infects B lymphocytes.
- Patients may manifest as a short-lived infectious mononucleosis / develop human cancers.
- The list of cancers produced include:
    - African form of Burkitt lymphoma.
    - B-cell lymphomas in immunosuppressed
    - A subset of Hodgkin lymphoma.
    - Nasopharyngeal carcinoma (T-cell tumor).
    - Some gastric carcinomas.
    - Rare forms of T-cell lymphomas and natural killer (NK) cell lymphomas.
    - Very rarely sarcomas.

PATHOGENESIS:

- EB virus infects B lymphocytes by binding to the membrane receptor CD21 (CR2).
- The infection of B-cells may be either productive (lytic) or latent.
- **Productive/lytic infection:** It develops only in a few patients and results in death of infected cells → release of virions → infection of other B-cells.
- **Latent infection:**
    - It occurs in majority of the cases.
    - The virus becomes latent inside the B-cells → are transformed or "immortalized" so that they are capable of proliferation indefinitely.
    - Immortalization of B lymphocyte is the hallmark of EBV infection.
    - Molecular basis of B-cell immortalization is related to two EBV-coded genes and viral cytokines.
        - ✓ **LMP1 (latent membrane protein 1):** It acts as an oncogene → activates the NF-κB and JAK/STAT signaling pathways → promote B-cell survival (prevents apoptosis by activating Bcl-2) and proliferation.
        - ✓ **EBNA2 (Epstein-Barr nuclear antigens 2):** It stimulates transcription of many host genes, including genes that drive the cell cycle (e.g. cyclin D) and the SRC family of proto-oncogenes.
        - ✓ **Viral cytokine (vIL-10):** It is pirated from host genome, prevent macrophages and monocytes from activating T-cells and killing viral infected cells



BURKITT LYMPHOMA:

- It is a B-cell neoplasm and is the most common childhood tumor in central Africa and New Guinea.
- A morphologically similar lymphoma occurs sporadically throughout the world.
- EBV is not directly oncogenic, but acts as a polyclonal B-cell mitogen → favors translocation → activate the c-MYC oncogene → release the cells from normal growth regulation.
- **Mechanism of endemic Burkitt lymphoma:** Normally, EBV infects B-cells and stimulate B lymphocyte proliferation which is controlled by suppressor T-cells.
- **Sequence of events in the pathogenesis of endemic African Burkitt lymphoma**:
  - Accompanying infections (such as malaria or other infections) impairs immune competence → lack of an adequate suppressor T-cell response → allows uncontrolled proliferation of B-cell.
  - EBV-infected B-cells expressing LMP-1 are eliminated by immune system.
  - Lymphoma develops only when there are chromosomal translocations that activate the c-MYC oncogene → results in uncontrolled proliferation of a malignant clone of B lymphocytes.
- **Mechanism of non-endemic Burkitt lymphoma:** All tumors possess the t (8; 14) or other translocations that dysregulate c-MYC.

NASOPHARYNGEAL CARCINOMA: 100% of nasopharyngeal carcinoma is associated with EBV.

HEPATITIS B AND C VIRUSES

- Do not encode any viral oncoproteins.
- Dominant effect -immunologically mediated chronic inflammation and hepatocyte death leading to regeneration and, over time, genomic damage.
- Immune cells produce mediators, such as reactive oxygen species, that are genotoxic and mutagenic.
- HBV and HCV genome contains genes that promote the development of cancer.

BACTERIAL CARCINOGENESIS:

HELICOBACTER PYLORI:

- Causes gastric adenocarcinomas and gastric lymphomas.
- Mechanism similar to HBV- and HCV. Increased epithelial cell proliferation in a background of chronic inflammation.
- H. pylori contains cytotoxin-associated A (CagA) gene → Initiates signaling cascade that mimics growth factor stimulation.
- Gastritis > gastric atrophy > Metaplasia > dysplasia > gastric adenocarcinoma.
- Gastric lymphomas are of B-cell origin.
- H. pylori infection leads to the appearance of H. pylori-reactive T cells, which in turn stimulate a polyclonal B-cell proliferation. – Monoclonal MALToma.

7. PATHWAYS OF SPREAD:

Invasiveness of cancers allows them to penetrate blood vessels, lymphatics and body cavities. It provides an opportunity for spread/dissemination of cancers through:

- Lymphatics
- Hematogenous
- Spread along body cavities
- Direct transplantation
- Epithelial lining

LYMPHATIC SPREAD:

- Most common pathway of spread for carcinomas.
- **Regional node involvement:** The walls of lymphatics in the region of cancer are readily invaded by cancer cells and form a continuous growth within the lymphatic channels (lymphatic permeation).
- Once the tumor cells gain access into the lymphatic vessels, they may detach to form tumor emboli and are carried to the regional draining lymph nodes.
- In the lymph node, the tumor emboli enter through afferent lymphatics at its convex surface and lodge and grow in the subcapsular sinus.
- Subsequently, the entire lymph node may be replaced by the metastatic tumor.
- Pattern of lymph node involvement follows the natural routes of lymphatic drainage.
- Sentinel lymph node biopsy is done to know the presence or absence of metastatic lesions.
- **Skip metastasis:**
  - When local lymph nodes are bypassed and lymphatic metastases develop in lymph nodes distant from the site of the primary tumor, these are called “skip metastasis”.
  - Example: Abdominal cancers may be first detected by an enlarged supraclavicular node such as Virchow’s node.
- **Retrograde metastasis:**
  - Tumors spreading against the flow of lymphatics may cause metastases at unusual sites.
  - Example: Carcinoma prostate metastasizing to supraclavicular lymph node.
- **Microscopic pattern of deposits:**
  - Initially, tumor cells are deposited in the marginal sinus and later extend throughout the node.
  - Micrometastases (microscopic involvement of lymph nodes) consist of single tumor cells or very small clusters.
- **Significance of lymph node metastases:**
  - Prognostic value, e.g. in breast cancer, involvement of axillary node is important for clinical diagnosis.

HEMATOGENOUS SPREAD:

- Hematogenous spread is usual in sarcomas but is also found in carcinomas.
- Blood borne metastasis usually occurs in **osteosarcoma, choriocarcinoma** and **renal cell carcinoma.**
- **Vessels invaded:** Cancer cells easily invade capillaries and venules, but thick-walled arterioles and arteries are resistant.
- **Tumors with affinity for venous invasion: Renal cell carcinoma**: It can invade the renal vein and grow in a snake-like fashion up the inferior vena cava, sometimes reaching the right side of the heart.

TARGET ORGAN FOR METASTASIS:

- **Liver and lungs**:
  - The most frequently involved organs; liver, because all portal area drains to the liver.
  - Tumors which penetrate systemic veins, eventually drain into the vena cava.
  - Since all caval blood flows to the lungs, it is the other common site for secondaries by hematogenous spread.
  - Through pulmonary veins, cancer cells from the primary lung cancer and metastatic deposit in the lungs may be carried to the left side of the heart.
  - From here the tumor emboli may be carried in systemic circulation to form secondary masses elsewhere in the body.
- **Bone metastasis**:
  - Cancer metastasizing to bone, prostate, lung, breast, liver, intestine, kidney and thyroid.
  - Bone metastasis may be
    - ✓ **Osteolytic lesion** which are characterized by radiolucencies
    - ✓ **Osteoblastic lesion** which are characterized by radiodensities
- **Other common sites:** Brain, lung, kidney and adrenals.
- **Organs relatively resistant:** skeletal muscle and spleen.

TRANSCELOMIC SPREAD:

- Malignant tumor arising in organs adjacent to body cavities (e.g. ovaries, gastrointestinal tract, and lung), may seed body cavities.
- The malignant cells may exfoliate or shed from the organ surfaces into the body cavities and cytological examination of this fluid may show malignant cells.
- Body cavities include peritoneal (most common), pleural cavities (common), pericardial (occasionally), joint space and subarachnoid space.

- **Peritoneal Cavity:** Example:
  - Ovarian Tumors, such as primary carcinomas of surface epithelial origin
  - Malignant GI tract tumors may spread to involve peritoneal cavity →ascites.
- **Pleural cavity:** Peripherally situated lung tumors → pleural effusions.
- **Cerebrospinal fluid:** Glioblastoma commonly spread through CSF into subarachnoid space to the spinal cord.

SPREAD ALONG THE EPITHELIAL LINED SPACES: (not common) Examples:

- Carcinoma endometrium may spread to ovary (or vice versa) through fallopian tube.
- Carcinoma of kidney may spread to lower urinary tract via ureters.

DIRECT TRANSPLANTATION: (rare) Tumor cells may be

- Directly transplanted (e.g.by surgical instruments like scalpel, needles, sutures) or
- Implantation by direct contact (e.g. transfer of cancer of lower lip to the corresponding opposite site in the upper lip).

SHORT ANSWERS

1. DIFFERENTIATION AND ANAPLASIA

DIFFERENTIATION:

- Differentiation refers to **the extent to which neoplastic parenchymal cells resemble the corresponding normal parenchymal cells, both morphologically and functionally**;
- This includes both morphological and functional differentiation.
- Differentiation determines the grade of the tumor

ANAPLASIA:

- Lack of differentiation (both structural and functional) is called as anaplasia
- It is characteristic of malignancy.
- The degree of anaplasia in a cancer cell correlates with the aggressiveness of the tumor.
- Thus, more anaplastic the tumor, the more aggressive it becomes.

MICROSCOPIC FEATURES OF ANAPLASIA:

- **Pleomorphism:**
  - It is defined as variation in the size and shape of cells and cell nuclei.
  - It is a feature of malignancy.
  - Thus, cells within the same malignant tumor range from large cells (many times larger than the neighbor cells), to extremely small (primitive appearing) cells.
- **Abnormal nuclear morphology:**
  - Extremely hyperchromatic nuclei of tumor cells are due to abundant chromatin and increased amount of DNA per cell compared to that of a normal cell.
  - Nuclear shape and size is variable and may be irregular.
  - Chromatin is coarsely clumped and distributed along the nuclear membrane.
  - Large prominent nucleoli are usually seen.
- **Mitoses:**
  - Presence of mitotic figures indicates the higher proliferative activity of the parenchymal cells.
  - **Number of mitotic figures:** Compared to benign and few well-differentiated malignant tumors, undifferentiated tumors usually show abundant (many) mitotic figures.
  - **Atypical (abnormal) mitotic figures:**
    - ✓ Normal mitosis produces bipolar spindles, and one cell divides into two.
    - ✓ When the mitotic spindles are more than two, it is called as atypical.
    - ✓ Presence of atypical bizarre mitotic figures is an important morphological feature of malignancy
  - **Nuclear cytoplasmic (N:C) ratio:**
    - ✓ In a normal cell, N:C ratio is 1:4 or 1:6.
    - ✓ In malignancy, the nuclei are enlarged, become disproportionately large for the cell
    - ✓ Hence, nuclear-to-cytoplasm ratio may be increased and may reach even up to 1:1.
- **Loss of polarity:**
  - Orientation of cells to one another is known as polarity.
  - The anaplastic cells lose the normal polarity markedly disturbed orientation (architecture) of tumor cells.
- **Growth pattern:**
  - Malignant neoplasms usually show disorganized growth.
  - The tumor cells may form sheets of cells, arranged around blood vessels, papillary structures, whorls, rosettes, etc.
  - Malignant tumors often show central ischemic necrosis due to compromised blood supply.
- **Bizarre cells, including tumor giant cells:** Some tumors may show bizarre cells with a single large polymorphic nucleus and others having two or more large, hyperchromatic nuclei.
- **Necrosis and apoptosis:** Many rapidly growing malignant tumors undergo large central areas of ischemic necrosis and/apoptosis.

RATES OF GROWTH:

Factors Determining the Rate of Growth:

- Degree of differentiation
- Dependency
- Balance between cell production and cell loss

Degree of differentiation

- Benign tumors are well-differentiated and usually grow slowly.
- Most malignant tumors grow more rapidly.

Dependency: Growth also depends on:

- Hormonal stimulation, e.g. uterine leiomyomas may suddenly grow during pregnancy and may undergo atrophy after menopause.
- Adequacy of blood supply.

Balance between cell production and cell loss: This in turn is determined by three main factors:

- **Doubling time of tumor cells**: It is the time required for the total cell cycle i.e. cell to double by mitosis.
- **Growth fraction:** It is the proportion of cells in the proliferative or replicative pool within the tumor.
- **Rate of tumor cell death:**
  - Rate of growth depends on balance between cell production and cell loss.
  - When both the rate of cell production and the rate of cell loss (by apoptosis) is high, it is termed as high cell turnover.

FUNCTIONAL CHANGES:

Well-differentiated tumors usually retain the functional characteristics. Function may be in the form of secretion and vary depending on the tumor type.

- **Secretion of normal substances:**
  - **Hormones:** Benign tumors and well-differentiated carcinomas of endocrine glands frequently secrete the hormones characteristic of their cell of origin (e.g. steroid hormones from an adrenocortical adenoma).
  - **Normal product:** Example: Well-differentiated squamous cell carcinomas produce keratin → form characteristic epithelial pearls.
- **Fetal proteins:**
  - Some tumors may secrete fetal proteins, which are not produced by comparable normal cells in the adult.
  - Example: Carcinoembryonic antigen (CEA) by adenocarcinomas of the GIT
- **Ectopic hormones:**
  - Tumors may produce substances which are not indigenous to the tissue of origin.
  - Example: Bronchogenic carcinomas may produce ACTH, parathyroid-like hormone, etc

2) ETIOLOGY OF CANCER:

AGE

Men – 60 to 79 years
Women- 40 to 79 years

ENVIRONMENTAL FACTORS

- Infectious agent E.g. human papilloma virus – cervical carcinoma
- Smoking
- Alcohol consumption
- Diet - Dietary factors increase the risk of colorectal prostate carcinoma
- Obesity
- Reproductive history - Long term exposure to estrogen stimulation – increase the risk of breast and endometrial cancer

ACQUIRED PREDISPOSING CONDITION

- **Chronic inflammation**
  - E.g. pancreatitis – pancreatic carcinoma
  - Barret esophagus- esophageal carcinoma
- **Precursor lesion**
  - Localized morphological changes associated with cancer risk. Eg:
  - Squamous metaplasia of bronchial mucosa in smoking
  - Colonic metaplasia in pernicious anemia
- **Immunodeficiency state**
  - Deficient T cell immunization- increase the risk for cancer caused by oncogenic virus

GENETIC PREDISPOSITION

Germline mutation in tumour suppressor gene – cause of inherited trait in families. E.g. BRCA1 AND BRCA2

Intrinsic risk factors	Non-intrinsic risk factors	
	Endogenous risk factors	Exogenous risk factors
❖ Random errors in DNA replication	❖ Biologic aging ❖ Genetic susceptibility ❖ DNA repair machinery ❖ Hormones ❖ Growth factors ❖ Inflammation ❖ etc.	❖ Radiation ❖ Chemical carcinogens ❖ Tumour causing viruses ❖ Bad lifestyles such as smoking, lack of exercise, nutrient imbalance ❖ etc.
[Unmodifiable]	[Partially modifiable]	[Modifiable]

3. METASTASIS:

DEFINITION:

Metastases are tumor deposits discontinuous with the primary tumor and located in a distant tissue. This process is known as metastasis and the resulting secondary deposits are called metastases.

SIGNIFICANCE:

- Metastases clearly identify a tumor as malignant because benign neoplasms never metastasize. Exceptions include two malignant tumors, which are locally invasive, but rarely metastasize.
    - Gliomas (malignant neoplasms of the glial cells) in the central nervous system.
    - Basal cell carcinomas of the skin.
- Metastases strongly reduce the possibility of cure of cancer.
- Metastatic spread is the most common cause of cancer death.

FACTORS FAVORING METASTASIS*:*

- Poorly differentiated tumor
- More rapidly growing tumor
- Large primary tumor.

Metastases: First important criteria for malignancy.

MORPHOLOGICAL APPEARANCE:

Microscopically, metastases resemble the primary tumor. But occasionally, they may be so anaplastic that their cell of origin cannot be made out.

Unknown primary: Sometimes metastases may appear without any clinically detectable primary tumor and the even microscopic examination of metastases may not reveal the characteristics features of primary site tumor.

Example: Metastases from adenocarcinoma may be so anaplastic that there is no evidence of any gland formation. In such situations, electron microscopic examination, immunohistochemistry by specific tumor markers will be helpful to establish the primary tumor.

4. PRECANCEROUS LESIONS:

Precancerous lesions are identifiable local signs (abnormalities) that, with time, have an increased risk of developing into cancer. Mostly these are skin lesions.

SOLAR KERATOSIS:

- Other name: Actinic, senile keratosis.
- Solar- sun induced keratosis.
- Lesions- mostly multiple lesions in sun exposed area.
- More common in fair skinned elder people.
- If untreated may lead to **Squamous Cell Carcinoma.**

MORPHOLOGY: Lesions are Tan brown, Erythematous.
HISTOLOGY: Hyperkeratosis, Acanthosis, Dyskeratosis Dysplasia

BOWEN'S DISEASE:

- Solitary lesions in sun exposed area.
- If left untreated may lead to Squamous cell Carcinoma.

MORPHOLOGY: Lesions are sharply circumscribed, rounded, reddish, brownish patches.
HISTOLOGY:

- Hyperkeratosis
- Epidermal Hyperplasia
- Dyskeratotic cells distributed through epidermis.

LENTIGO MALIGNA:

- Melanocytes become malignant and grow continuously over the Stratum basale.
- If untreated may lead to Melanoma.

MORPHOLOGY: Blue/Black stain
HISTOLOGY: Epidermal Atrophy, Increased Melanocytes.

5. p53:

TP53 GENE (GUARDIAN OF THE GENOME)

- TP53 is a tumor suppressor gene located on small arm of chromosome 17.
- Loss-of-function mutations in TP53 is the most common mutations observed in more than 50% of cancers.

FUNCTIONS OF p53:

- TP53 sense cellular stress, DNA damage, shortened telomeres, hypoxia and stress caused due to increased pro-growth signaling (e.g. cells with mutations in RAS and MYC genes).
- In healthy cells, p53 is maintained at low levels by MDM2 (degrader of p53)
- In stressed cells, p53 is released from the inhibitory effects of MDM2 and p53 becomes activated.

ACTIVATION OF p53:

Stress due to DNA damage or hypoxia activates

- ataxia-telangiectasia mutated (ATM) and
- ataxia-telangiectasia and Rad3 related (ATR).

Activated ATM and ATR stimulate the phosphorylation of p53 and MDM2. This disrupts the binding and degradation of p53 by MDM2 and leads to activation and accumulation of p53.

Oncogenic stress: By oncoproteins such as RAS cellular stress and lead to increased expression of p14/ARF. p14/ARF binds MDM2 and releases p53 and resulting in raised p53 levels in the cell.

PREVENTION OF NEOPLASTIC TRANSFORMATION:

Activated p53 prevents neoplastic transformation of cell by three interconnected mechanisms:

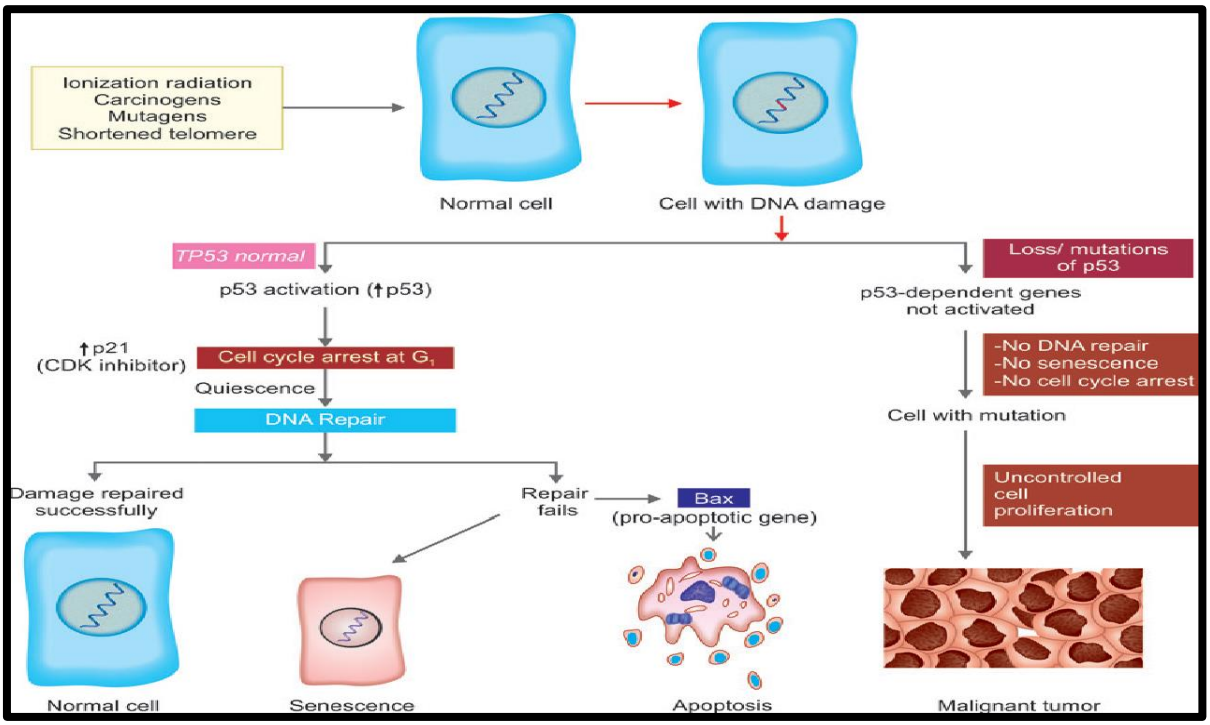
- **Transient/temporary p53-induced cell cycle arrest:**
  - If there is damage to DNA, transient, rapid cell cycle arrest occurs late in the G1 phase.
  - It is brought-out partly by p53-dependent transcription of the CDKN1A gene (encodes the CDK inhibitor p21).
  - p21 in turn inhibits CDK4/D cyclin complexes and maintain RB in an active, hypophosphorylated state.
  - This blocks the progression of cells from G1 phase to S phase. This cell cycle arrest gives the cells time to repair DNA damage.
  - If DNA damage is repaired, the signals that caused stabilization/ activation of p53 disappears.
  - This results in fall in the levels of p53 and releases the block in cell cycle and return of cells to a normal state

- **p53-induced senescence (permanent cell cycle arrest):**
  - Senescence is defined as a state of permanent cell cycle arrest.
  - Senescence may be stimulated in response to different types of stresses (e.g. unopposed oncogene signaling, hypoxia and shortened telomeres).
  - The senescent cells are prevented from forming tumors.
- **p53-induced apoptosis (programmed cell death):**
  - Cells with irreversible DNA damage undergo p53- induced apoptosis and is the protective mechanism against development of cancer.
  - p53 stimulates transcription of several pro-apoptotic genes (e.g. BAX and PUMA) resulting in apoptosis of cells via the intrinsic (mitochondrial) pathway.

METHOD OF INACTIVATION OF TP53 GENE AND ASSOCIATED TUMORS:

Most cancers have defect in TP53 gene.

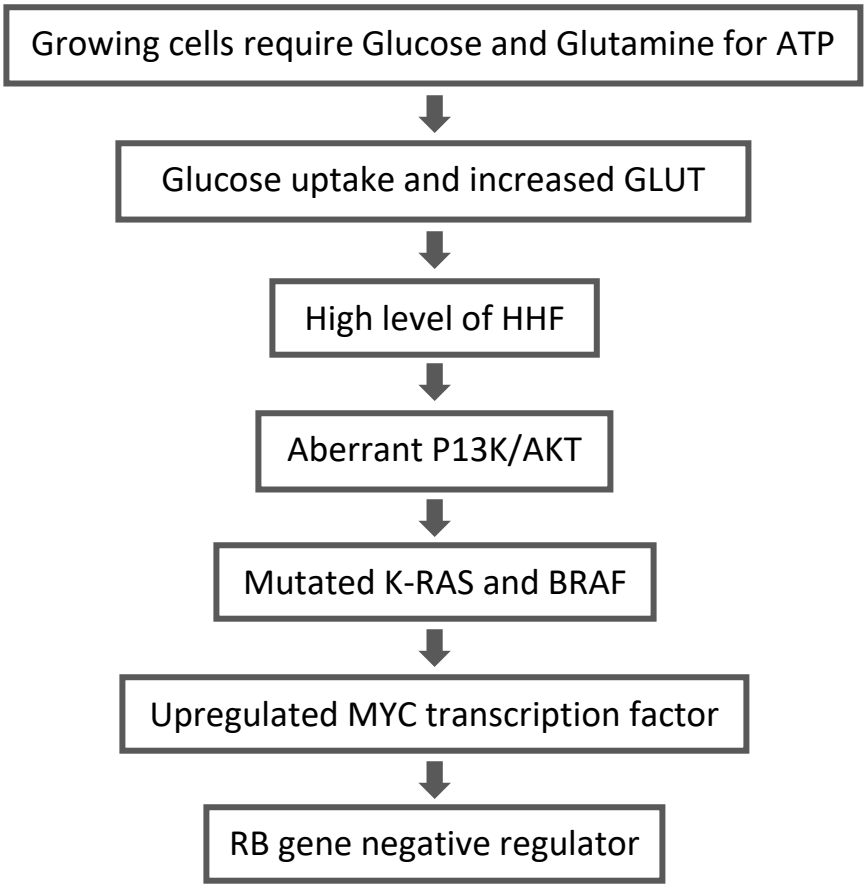
- **Acquired loss-of-function mutation in both (biallelic)** TP53 alleles in somatic cells is most common.
- **Germ-line mutations in one TP53 allele:** It is less common. Individuals may inherit one mutated/defective **TP53** allele and one additional "hit" in the other normal **TP53** allele will produce malignant tumors. For example: Li-Fraumeni syndrome
- **Mutations of proteins that regulate p53 function:** MDM2 and related proteins of the MDM2 (enzyme that ubiquitinylates p53) family degrade p53 leading to a functional deficiency of p53. (Overexpressed)
- **Blocking of p53 function**: Transforming proteins of many DNA viruses bind and degrade p53 even without mutation in p53. Eg: Viral oncoprotein E6 of high-risk human papillomaviruses (HPVs)



6. WARBURG EFFECT:

Cancer cells have some change in their metabolism, so that they promote initiation and progression of tumors.

EXCESSIVE NUTRIENT ACQUITION:



OTHERS:

- Entosis
- Phagocytosis
- Macropinocytosis.

ALTERED METABOLIC PATHWAY:

- Mitochondrial oxidative phosphorylation
- Kreb's-citric acid cycle
- Glutamine converts to Glutamate.

ONCOMETABOLITES IN TUMOUROGENESIS:

2HG (Hydroxyglutamate)-inhibits dioxygenase Eg: Triple negative breast cancer
Metabolites that regulates Epigenetic phenomena of acetylation and methylation.

DYNAMIC TUMOR MICROENVIRONMENT:

- Angiogenic vascular cells
- Cancer associated fibroblasts cells
- Infiltrating immune cells

EVADING IMMUNE DESTRUCTION:

- Host immunity against non-self-cells

TUMOR ANTIGEN:

- Oncoproteins from mutated oncogenes E.g.: RAS, BCR/ALB, CDK4
- Protein products of Tumor suppressor genes: E.g.: P53, Beta catenin
- Overexpressed cellular protein: HER2
- Abnormally expressed cellular protein: MAGE, GAGE
- Tumor antigen from viral oncoprotein: ENBA protein
- Tumor antigen from randomly mutated gene:
- Cell specific differentiation antigen:
- Prostate Specific Antigen
- Oncofetal antigen
- Alpha fetoprotein
- Abnormal cell surface molecule: MUC1 in breast cancer.

ANTI-TUMOR IMMUNE RESPONSE:

- Cell mediated immunity by
- Cytotoxic T cells
- NK cells
- Macrophages
- Humoral immunity by antibodies.

ESCAPE FROM IMMUNE SURVEILLANCE:

- Cancer immunomodelling - immune system changes, shapes the malignant cells.
- It eliminates cancer cells, develops equilibrium between cancer cells and immune competence to eliminate tumor cells

7. GROWTH FACTORS:

- Growth factors are polypeptides that stimulate the activity of proteins that are required for cell survival, growth and division.
- MOA: Growth factor activity is mediated through binding to specific receptors, ultimately influencing the expression of genes that can:
- Promote entry of cells into the cell cycle
- Relieve blocks on cell-cycle progression (thus promoting replication)
- Prevent apoptosis
- Enhance biosynthesis of cellular components (nucleic acids, proteins, lipids, carbohydrates) required for a mother cell to give rise to two daughter cells.

EXAMPLES OF GF:

Table 1.1 Growth Factors Involved in Regeneration and Repair

Growth Factor	Sources	Functions
Epidermal growth factor (EGF)	Activated macrophages, salivary glands, keratinocytes, and many other cells	Mitogenic for keratinocytes and fibroblasts; stimulates keratinocyte migration; stimulates formation of granulation tissue
Transforming growth factor-α (TGF-α)	Activated macrophages, keratinocytes, many other cell types	Stimulates proliferation of hepatocytes and many other epithelial cells
Hepatocyte growth factor (HGF) (scatter factor)	Fibroblasts, stromal cells in the liver, endothelial cells	Enhances proliferation of hepatocytes and other epithelial cells; increases cell motility
Vascular endothelial growth factor (VEGF)	Mesenchymal cells	Stimulates proliferation of endothelial cells; increases vascular permeability
Platelet-derived growth factor (PDGF)	Platelets, macrophages, endothelial cells, smooth muscle cells, keratinocytes	Chemotactic for neutrophils, macrophages, fibroblasts, and smooth muscle cells; activates and stimulates proliferation of fibroblasts, endothelial, and other cells; stimulates ECM protein synthesis
Fibroblast growth factors (FGFs), including acidic (FGF-1) and basic (FGF-2)	Macrophages, mast cells, endothelial cells, many other cell types	Chemotactic and mitogenic for fibroblasts; stimulates angiogenesis and ECM protein synthesis
Transforming growth factor-β (TGF-β)	Platelets, T lymphocytes, macrophages, endothelial cells, keratinocytes, smooth muscle cells, fibroblasts	Chemotactic for leukocytes and fibroblasts; stimulates ECM protein synthesis; suppresses acute inflammation
Keratinocyte growth factor (KGF) (i.e., FGF-7)	Fibroblasts	Stimulates keratinocyte migration, proliferation, and differentiation

GROWTH FACTORS IN TUMORIGENESIS:

- Growth factors can be involved in the proliferation of cells at steady state as well as after injury, when irreversibly damaged cells must be replaced.
- Uncontrolled proliferation can result when the growth factor activity is dysregulated, or when growth factor signaling pathways are altered to become constitutively active.
- Thus, many growth factor pathway genes are proto-oncogenes, and gain of function mutations in these genes can convert them into oncogenes capable of driving unfettered cell proliferation and tumor formation.

TABLE 7-6 Selected Oncogenes, Their Mode of Activation, and Associated Human Tumors

Category	Proto-oncogene	Mode of Activation	Associated Human Tumor
GROWTH FACTORS			
PDGF-β chain	*SIS* (official name *PBGFB*)	Overexpression	Astrocytoma Osteosarcoma
Fibroblast growth factors	*HST1* *INT2* (official name *FGF3*)	Overexpression Amplification	Stomach cancer Bladder cancer Breast cancer Melanoma
TGF-α	*TGFA*	Overexpression	Astrocytomas Hepatocellular carcinomas
HGF	*HGF*	Overexpression	Thyroid cancer
GROWTH FACTOR RECEPTORS			
EGF-receptor family	*ERBB1* (*EGFR*), *ERRB2*	Overexpression	Squamous cell carcinoma of lung, gliomas
FMS-like tyrosine kinase 3	*FLT3*	Amplification	Breast and ovarian cancers
Receptor for neurotrophic factors	*RET*	Point mutation Point mutation	Leukemia Multiple endocrine neoplasia 2A and B, familial medullary thyroid carcinomas
PDGF receptor	*PDGFRB*	Overexpression, translocation	Gliomas, leukemias
Receptor for stem cell (steel) factor	*KIT*	Point mutation	Gastrointestinal stromal tumors, seminomas, leukemias

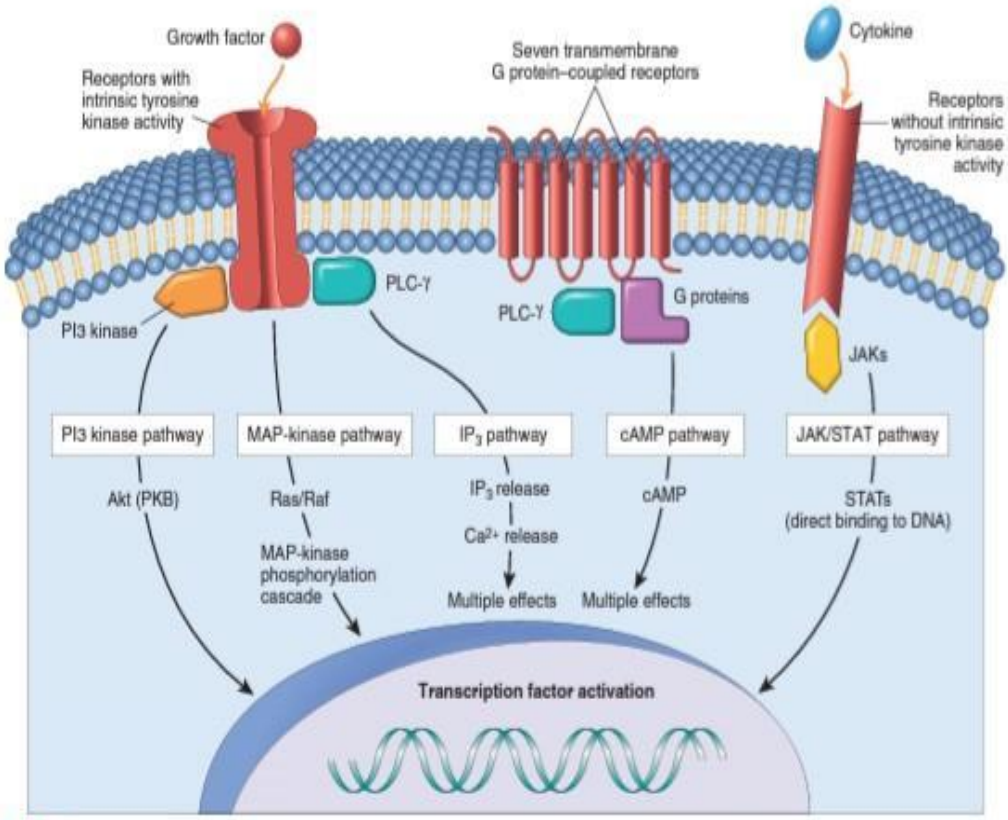


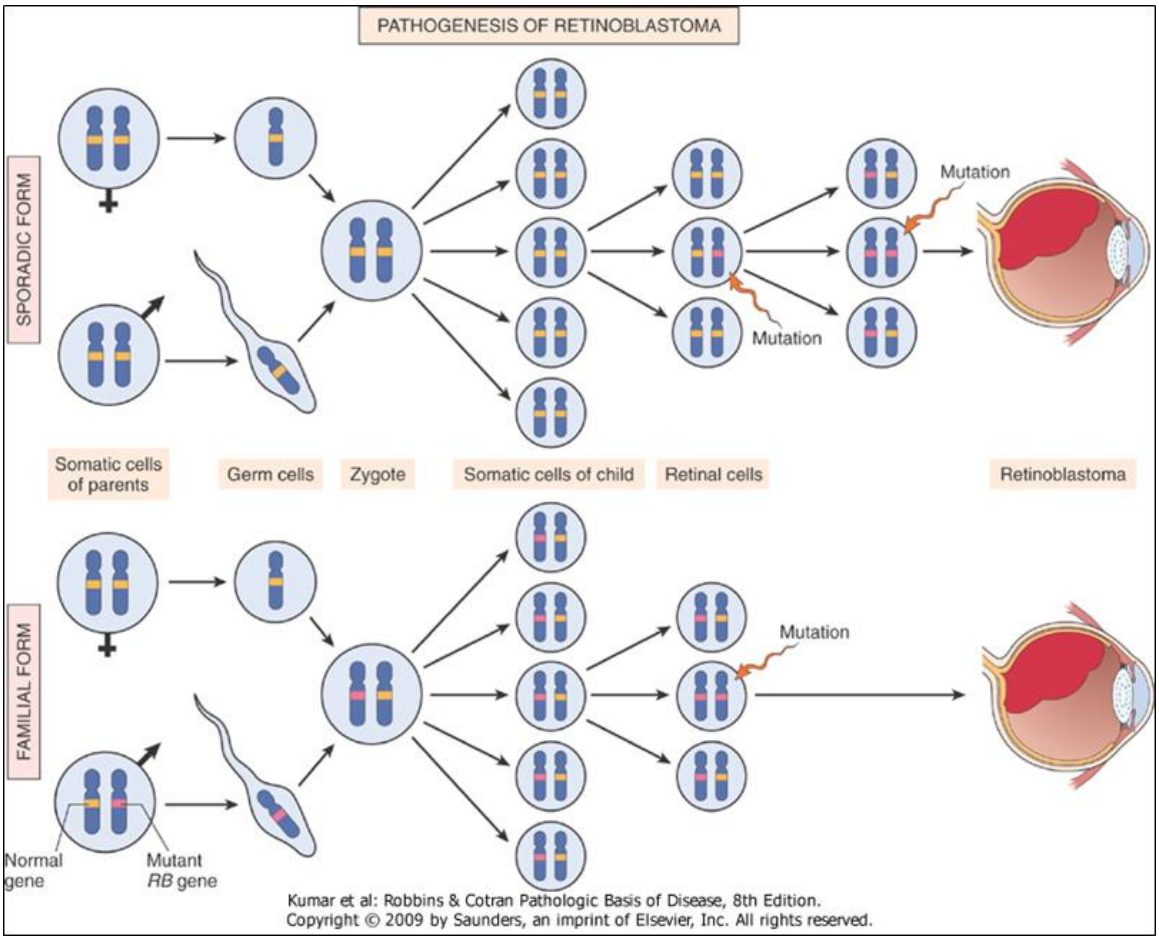
FIGURE 3–9 Overview of the main types of cell surface receptors and their principal signal transduction pathways (see text). Shown are receptors with intrinsic tyrosine kinase activity, seven transmembrane G protein–coupled receptors, and receptors without intrinsic tyrosine kinase activity. cAMP, cyclic adenosine monophosphate: IP_3, inositol triphosphate; JAK, Janus kinase; MAP kinase, mitogen-activated protein kinase; PI3 kinase, phosphatidylinositol 3-kinase; PKB, protein kinase B, also known as Akt; PLC-γ, phospholipase C gamma; STATs, signal transducers and activators of transcription.

8. RETINOBLASTOMA:

RETINOBLASTOMA GENE (RB GENE)

- RB (RB1) gene was the first discovered tumor suppressor gene, which is present on **chromosome locus 13q14**
- Inactivation of RB gene was found in retinoblastoma.
- Retinoblastoma - hereditary or sporadic form.

KNUDSON'S TWO-HIT HYPOTHESIS OF RETINOBLASTOMA:



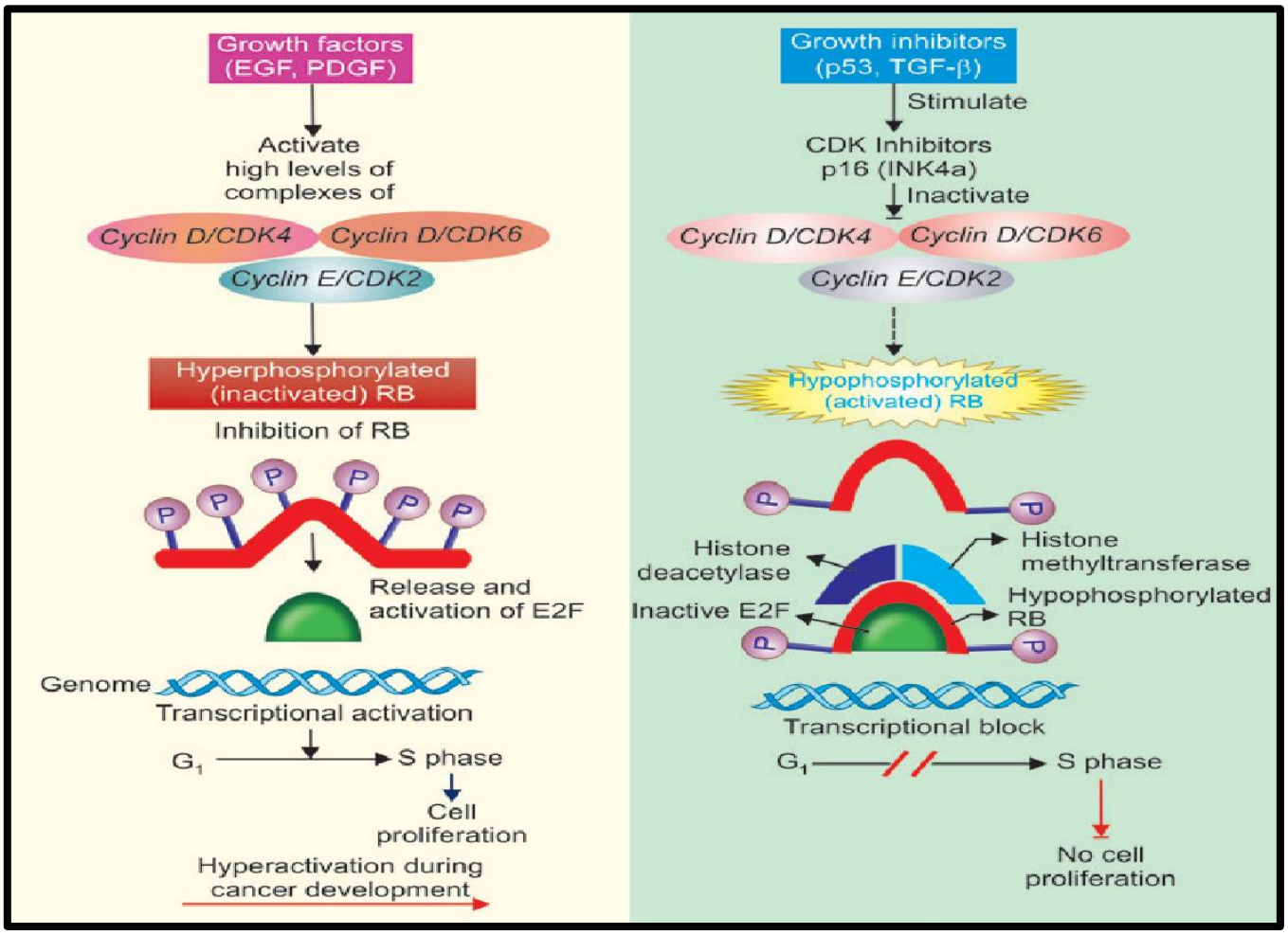
Two mutations (hits), involving both alleles of tumor suppressor gene are required to produce the tumor.

- In familial cases, one mutation (first hit) takes place in the germ line and second hit after birth. (40%)
- In sporadic cases both mutations (two hits) develop after birth (60%)

FUNCTIONS OF THE RB GENE:

RB gene cell cycle regulation & cellular differentiation.

Active RB gene regulates G1/S checkpoint of cell cycle



- Loss of function mutation in both RB alleles
- A shift from active hypophosphorylated state to the inactive hyperphosphorylated state by gain-of function mutations that upregulate CDK / cyclin D activity or by loss-of-function mutations of CDK inhibitors(p16)
- Human papilloma virus (HPV) - E7 protein bind to hypophosphorylated RB and release E2F transcription factor - Cervical carcinoma

Function of RB in regulating the G1-S checkpoint of the cell cycle:

- When RB is phosphorylated by the cyclin D–CDK4/6 complexes, it releases E2F. The latter then activates transcription of S-phase genes;
- Hypophosphorylated active RB combines with the E2F transcription factors along with histone deacetylases and histone methyltransferases, and inhibits progression from G1-S phase of cell cycle

9. TUMOR MARKERS:

TUMOR MARKERS	TUMOR TYPES
Human chorionic gonadotropin	Trophoblastic tumors, nonseminomatous testicular tumors **(choriocarcinoma)**
Calcitonin	Medullary carcinoma of thyroid
Catecholamines and metabolites	Pheochromocytoma and related tumors
Alpha fetoprotein	Liver cell cancer, nonseminomatous germ cell tumors of testis **(yolk sac tumor)**
Carcinoembryonic antigen	Carcinomas of Colon, pancreas, lung, stomach, heart
Prostatic acid phosphatase	Prostate cancer
Neuron specific enolase	Small cell cancer of lung, neuroblastoma
Immunoglobulins	Multiple myeloma and other gammopathies
Prostate specific antigen Prostate specific membrane antigen	Prostate cancer
CA-125	Ovarian cancer
CA-19-9	Colon Cancer, pancreatic cancer
CA-15-3	Breast cancer
TP53, APC, RAS mutants in stool and serum	Colon cancer
TP53, RAS mutants in stool and serum	Pancreas cancer
TP53, RAS mutants in sputum and serum	Lung cancer
TP53 mutants in urine	Bladder cancer

10. EXFOLIATIVE CYTOLOGY:

- Exfoliated (shed) cells - evaluated for anaplastic features.
- Used to differentiate normal, malignant, and dysplastic cells.
- Limitations:
  - Poorly differentiated tumor and some specific tumor types are difficult to distinguish based on their morphological appearance.
- Uses: to evaluate
  - Cervical cancer
  - Lung carcinoma
  - Endometrial carcinoma
  - Bladder and prostatic carcinoma
  - Gastric carcinoma
  - Identify tumor cells in CSF, joint, abdominal, pleural fluid

11. ONCOFOETAL PROTEINS:

ONCOFOETAL ANTIGENS:

- E.g. alpha fetoprotein (AFP), Carcinoembryonic antigen (CEA)
- Expressed at higher levels in fetal tissues and cancer cells [not limited to tumors, increased during inflammations, small quantities even in normal tissues
- Aid in tumor diagnosis and clinical management
- AFP – Liver cancer; CEA – Colon cancer

IMMUNE SURVEILLANCE:

- Coined by Lewis Thomas and Macfarlane Burnet
- Normal function of the immune system to constantly scan the body for any emerging malignant cells and destroy them.

CLINICAL OBSERVATIONS SUPPORTING IMMUNE SURVEILLANCE:

- **Immune evasion by cancers:**
- **Cancer immunoediting:** Process in which immune system
  - Eliminates cancer cells
  - Develop equilibrium with cancer cells or
  - Escape of cancer cells.
- **ESCAPE MECHANISMS:**
  - Selective outgrowth of antigen – negative variants.
  - Loss or reduced expression of MHC molecules.
  - Activation of immunoregulatory pathways.
  - Secretion of immunosuppressive factors by cancer cells.
  - Induction of regulatory T cells.

12. LABORATORY DIAGNOSIS OF CANCER:

TISSUE SAMPLING:

- Sampling must be adequate
- Appropriate preservation (formalin solution)
- Based on differential diagnosis tissue should be rapidly allocated for other diagnosis (cytogenetics, flow cytometry, molecular diagnosis)
- Sampling approaches
  - Excision /biopsy
  - Needle aspiration.
  - Cytologic smears

FLOW CYTOMETRY:

- Measures individual cell characteristics
- Multiple antigens can be assessed simultaneously using different combinations of antibodies with different fluorescent dyes.
- Uses: identification of B- and T- cell lymphoma, leukemia, myeloid neoplasm.

LIQUID BASED CYTOLOGY:

- Special technique for preparation of samples- uniform monolayered dispersion of cells on smears.

CIRCULATORY TUMOR CELLS:

- Detection, quantification, and characterization of rare solid tumors (carcinoma, melanoma) circulating in the blood with the help of 3-D flow cells coated with specific antibodies.

13. FNAC: Fine Needle Aspiration Cytology

PROCEDURE:

- Small bore needle
- Cells and attendant fluid aspirated
- Stained smears prepared
- Microscopic examination

SITE OF ASPIRATION:

Readily palpable superficial lesions

- Breast
- Thyroid
- Lymph node

Deep seated structures

- Pelvic lymph nodes
- Pancreas

ADVANTAGES:

- Less invasive, rapidly performed.
- Extremely reliable and useful
- Prevents surgery and associated risk

14. AUTOMATIC TISSUE PROCESSOR:

- A machine routinely used in the histopathology laboratory.
- A tissue processor is a device that prepares tissue samples for sectioning and microscopic examination in the diagnostic laboratory.
- Microscopic analysis of cells and tissues requires the preparation of very thin, high quality sections (slices) mounted on glass slides and appropriately stained to demonstrate normal and abnormal structures.
- The ATP machine plays a role in the preparation of the tissue by passing them through various chemicals; a major process called tissue processing.

STEPS OF TISSUE PROCESSING:

Tissues to be processed are cut into small pieces to ensure the tissue fits into the tissue cassettes. Smaller tissues (2-4 um) will be processed faster than the whole tissue or organ. These tissue cassettes are packed into the oscillating tissue basket to tissue prior to fixation.

- **Fixation**
  - This is the process of preserving or fixing tissues by passing them through chemicals called fixatives.
  - The fixatives will help protect the tissue from decay and autolysis.
  - Routine fixative of use is 10% formalin
- **Dehydration**
  - This is the process of removing water molecules from the tissue by passing the tissue through ascending grades of alcohol.
  - E.g methanol, acetone, 70-100% alcohol
- **Clearing**
  - This is the process of removing alcohol from the tissue by passing it through chemicals that will remove the alcohol molecules.
  - These agents are called clearing agents.
  - Xylene is mostly used for clearing.
- **Infiltration**
  - This is the process of filling intracellular spaces left in the tissue by paraffin wax.
  - This will help confer a bit of rigidity to the processed tissue.
- **Embedding**
  - This last step is manually done.
  - This has to do with immersing the processed tissue into a mould containing liquid paraffin wax.
  - This is for external support so that the tissue won't crumble during microtomy

PARTS OF THE ATPM:

- Oscillating tissue basket
- 10 beakers or jars- The transparent beakers is for processing fluids
- 2 thermostatically controlled beakers-The beaker is for parrafin wax
- An electric rotor at the base
- Lifting mechanism
- Time disc and alarm system
- Control unit - with display screen and control buttons

WORKING PRINCIPLE OF ATPM:

- The tissue basket oscillates up and down in each station at three-second intervals to ensure thorough and even mixing of the reagents and optimum tissue infiltration.
- Infiltration time is separately programmable for each station.
- Up to nine programs may be run with immediate or delayed starting times.
- When it's time for tissue to be transferred to the next beaker or jar, the cover of the machine is raised up, and the lifting mechanism carefully removes the tissue basket and gently transfers it to the next beaker.
- When the infiltration time for any particular station is exceeded, a warning message will display, indicating the station number and excess time.
- Controls are arranged by functionality with an LCD to indicate operational parameters.
- Reagent container lids have seals to minimize operator exposure to hazardous fumes.
- Tissue basket immediately immerses in a station in the event of power loss to protect samples from drying out.
- When power is restored, program will resume.
- In the event of long-term power failure, wax is liquified.
- Capacity of tissue basket is 80 cassettes.
- Vacuum configurations hasten infiltration, allowing pressure to be applied to any station in either manual or automatic operation.
- Fume control configurations extract fumes with a fan and pass them through an internal carbon filter.
- For added efficiency, these models feature a two-part containment shield surrounding the reagent container platform.

ADVANTAGES OF ATPM:

- It's very efficient
- Saves time and energy to operate
- Cost effective and user friendly
- Can process different tissues same time
- The machine does the transfer of tissue from one bath to another.

ATPM – PROCESSING TIME SCHEDULE:

Processing schedule varies and it depends on the following:

- Nature and size of tissue
- Urgency

- Beaker I – fixative (formalin) 1-2 hours
- Beaker II – fixative 1 hour
- Beaker III – fixative. 30- 45 minutes
- Beaker IV – 70% alcohol. 30 minutes
- Beaker V – 90% alcohol. 30 minutes
- Beaker VI – Absolute alcohol. 1 hour
- Beaker VII – Absolute alcohol. 1 hour
- Beaker VIII – Methanol 30 minutes
- Beaker IX – Xylene. 1-2 hours
- Beaker X – Xylene 45 minutes – 1 hour
- Wax bath I (done at 45°c) 2 hours
- Wax bath II. 2 hours

All the processes are performed by the ATPM from start to finish except embedding which is best done manually.

15. PARANEOPLASTIC SYNDROME:

- Signs and symptoms that cannot readily be explained by the Anatomic distribution of the tumor or by the elaboration of hormones indigenous to the tissue from which the tumor arose
- Most common: **Hypocalcaemia.**

EXAMPLES:

CUSHING SYNDROME	Small cell carcinoma of lung Pancreatic carcinoma.
HYPERCALCEMIA	Squamous cell carcinoma of lung Breast carcinoma Renal carcinoma
ACATHOSIS NIGRICANS	Gastric carcinoma Lung carcinoma
VENOUS THROMBOSIS	Pancreatic carcinoma Bronchogenic carcinoma
DISSEMINATED INTRAVASCULAR COAGULATION	Acute promyelocytic leukemia Prostatic carcinoma

16. GRADING AND STAGING OF TUMORS:

GRADING:

- Based on degree of differentiation of tumor cells
- Determined by cytological appearance
- High grade tumors- anaplastic (e.g. Pancreatic adenocarcinoma)
- Low grade tumors- well differentiated (e.g. adenocarcinoma of stomach)

STAGING:

Determined by surgical exploration/imaging
Has greater clinical value than grading

TNM STAGING SYSTEM:

T-primary tumor.
    T0-in situ carcinoma
    T1-T4--increasing size
N-regional lymph node involvement
    N0-no lymph node involved
    N1-N3-- increasing no of lymph node involved
M-metastases
    M0-no distant metastases
    M1-M2--metastases present

17. CARCINOMA IN-SITU

A pre invasive stage of cancer is called as carcinoma in-situ

DEFINITION:

Carcinoma in situ is defined as:

- A preinvasive epithelial neoplasm.
- Shows all the cytological features of malignancy
- Involves the entire thickness of the epithelium.
- Remains confined within the epithelial basement membrane.

The tumor cells cannot reach the potential routes of metastasis, such as blood vessels and lymphatics until the basement membrane has been breached or invaded.

SHORT ANSWERS

1. IMMUNE EVASION BY CANCERS:

CANCER IMMUNOEDITING:

Process in which immune system

- Eliminates cancer cells,
- Develop equilibrium with cancer cells or
- Escape of cancer cells.

ESCAPE MECHANISMS:

- Selective outgrowth of antigen – negative variants.
- Loss or reduced expression of MHC molecules.
- Activation of immunoregulatory pathways.
- Secretion of immunosuppressive factors by cancer cells.
- Induction of regulatory T cells.

2. DYSPLASIA

It literally means disordered growth. The changes of dysplasia include:

- Cellular pleomorphism.
- Large hyperchromatic nuclei.
- High nuclear-to-cytoplasmic ratio.
- Loss of polarity (architectural orientation).

Classification of dysplasia:

- Mild,
- Moderate
- Severe.

] depending on the thickness of epithelium involved by the dysplastic cells

FATE:

- Mild-to-moderate dysplastic changes, which do not involve the entire thickness of epithelium may be reversible, if the cause is removed.
- Thus, dysplasia need not progress to cancer.
- Once the tumor cells breach the basement membrane, the tumor is said to be invasive and carcinoma in situ may take years to become invasive.
- Most in situ tumor, with time penetrate the basement membrane and invade the subepithelial stroma.

SITES:

- Uterine cervix, skin and breast.
- Asymptomatic: In this stage, tumors are usually asymptomatic.

3. METAPLASIA:

- It is reversible change in which one type of differentiated cell is replaced by another type of differentiated cells.
- It is a cellular adaptation that develops in association with tissue damage, repair and regeneration.
- Examples: Gastroesophageal reflux damages the squamous epithelium of the esophagus which is replaced by glandular (gastric or intestinal) epithelium,
- Columnar epithelium of endocervix is replaced by stratified squamous epithelium.
- Malignancy may develop in these metaplastic epithelia.

4. HAMARTOMA:

- Mixed type of tumor.
- Hamartomas are disorganized but benign masses composed of cells indigenous to the involved site.
- Once thought to be a developmental malformation unworthy of the – oma designation.
- Many in fact have clonal chromosomal aberrations that are acquired though somatic mutations and on basis are now considered neoplasm.

5. TERATOMA

- The great majority of neoplasms, even mixed tumors, are composed of cells from a single germ layer.
- An exception is a tumor called teratoma, which contains recognizable **mature or immature cells or tissues** belonging to **more than one germ layer (and sometimes all three).**
- Teratoma originates from totipotential germ cells that are normally present in the ovary and testis and sometimes also found in abnormal midline embryonic rests.
- Such cells can differentiate into any of the cell types found in the adult body and so, not surprisingly, may give rise to neoplasms that contain, in a helter-skelter fashion, bone, epithelium, muscle, fat, nerve, and other tissues.
- A particularly common pattern is seen in the ovarian cystic teratoma (dermoid cyst)

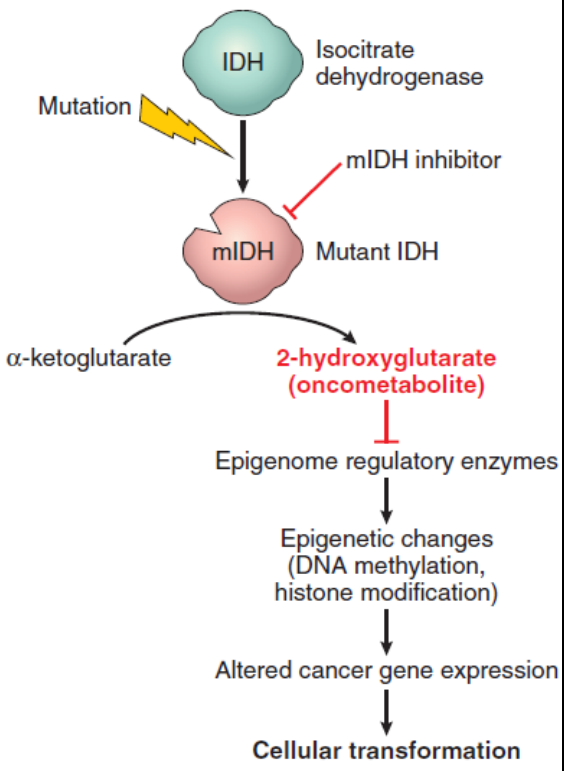
6. FROZEN SECTION:

- Tissue is frozen and sections are cut using freezing microtome/cryostat.
- Highly accurate
- Uses:
  - Rapid diagnosis
  - Determine the nature of a mass lesion
  - Evaluation of a margin of an excised lesion

UPDATES:

1. ONCOMETABOLISM

- Tumor genome sequencing studies has revealed mutations in enzymes that participate in the Krebs cycle. Of these, mutations in isocitrate dehydrogenase (IDH) have revealed a new mechanism of oncogenesis termed oncometabolism
- The proposed steps in the oncogenic pathway involving IDH are as follows:
  - IDH → mutation → amino acid substitution in residues at active site → loss of IDH function + new enzymatic activity that catalyzes the production of 2-hydroxyglutarate (2-HG) → inhibits enzymes that require α-ketoglutarate as a cofactor.
  - Among the proteins inhibited by 2-HG (oncometabolite) are several members of the TET family, including TET2.
  - TET2 enhances DNA methylation, which controls normal gene expression.
  - Loss of TET2 activity → Abnormal DNA methylation → mis-expression of cancer genes → cellular transformation and oncogenesis.
  - Net effect of TET2 loss is the upregulation of RAS and receptor tyrosine kinase signaling.



- Oncogenic IDH mutations occur in cholangiocarcinomas, gliomas, acute myeloid leukemias, and sarcomas.
- Newer drugs that inhibit mutated IDH and not the normal IDH enzyme are now being used to treat leukemias with IDH mutations.

2. PRECURSOR LESIONS.

- Precursor lesions are defined by localized morphologic changes that identify a field of epithelium that is at increased risk for malignant transformation.
- These changes may take the form of hyperplasia, metaplasia, or dysplasia.
- Precursor lesions consisting of hyperplasia often stem from chronic exposure to trophic factors.

3. METASTATIC PHENOTYPE:

- At each point in metastatic cascade, the breakaway cells should overcome the challenges of avoiding immune defences and adapting to a microenvironment (e.g., lymph node, bone marrow, or brain)
- The complexity of this series of events may explain why individual "metastasis genes" have not been found; it may be that the "metastatic phenotype" requires the accumulation complementary genetic and epigenetic alterations that collectively promote the metastatic cascade.
- This set of "skills" may be present only in rare cells, or might even require collaboration between subclones, with each providing some needed function.
- This latter idea implies that successful metastases may arise from cells that migrate as cohesive groups.

4. EPITHELIAL MESENCHYMAL TRANSITION

- It is hypothesized that E-cadherin expression is silenced through a process called epithelial-mesenchymal transition (EMT) - an integral to the metastasis of carcinomas, particularly breast and prostate cancers.
- EMT is controlled by the transcription factors SNAIL and TWIST
- It is defined by:
  - Downregulation of epithelial markers (e.g., E-cadherin)
  - Concomitant upregulation of mesenchymal markers (e.g., vimentin and smooth muscle actin)

    changes that are believed to favour the development of a promigratory phenotype that is essential for metastasis.

5. LOSS OF p53 FUNTION ANF GENOMIC INSTABILITY:

- Cancers with loss of p53 function not only accumulate point mutations but also are strongly associated with aneuploidy, which may take the form of deletions, amplifications, and complex chromosomal rearrangements.
- These genomic aberrations may occur in cells with defective telomeres during break fusion-breakage cycles or may be created by other types of chromosomal "catastrophes" that lead to DNA breaks in multiple chromosomes.

6. ROLE OF CD44 ADHESION MOLECULE

- CD44 adhesion molecule is expressed on normal T lymphocytes.
- It is used by tumour cells to migrate to selective sites in lymphoid tissues.
- Such migration is accomplished by the binding of CD44 to hyaluronate on high endothelial venules.
- CD44, which appears to enhance spread of tumour to lymph nodes and other metastatic sites

7. ANOIKIS – (meaning: without a home)

- Loss of adhesion in normal cells leads to induction of apoptosis, but free tumor cells are resistant to this form of cell death
- This is due to the expression of integrins that mitigate the loss of adhesion to ECM, apparently by transmitting signals that promote cell survival.

8. MALIGNANT NATURE OF WELL AND POORLY DIFFERENTIATED TUMOUR

- Malignant nature of well-differentiated tumours is revealed by the invasion of adjacent tissues and their ability to metastasize.
- At the other end of the spectrum is highly anaplastic, poorly differentiated tumours exhibiting little or no evidence of differentiation - a morphologic appearance that is highly predictive of malignant behaviour.

9. EXTRAVASATION OF TUMOUR CELLS

- Once arrested at distant sites, extravasation of tumour cells involves transmigration between endothelial cells followed by egress through the basement membrane.
- Extravasation requires the action of adhesion molecules (integrins, laminin receptors), proteolytic enzymes, and chemokines, which may be derived from tumour cells or from innate immune cells such as monocytes and neutrophils.

10. METAPLASIA, DYSPLASIA, AND CARCINOMA IN SITU:

These terms describe morphologically recognizable changes in differentiation that represent:

- Adaptation to chronic injury (metaplasia)
- Premalignant change (dysplasia),
- Cancer that has yet to invade (carcinoma in situ) – aka preinvasive neoplasm

11. ALK

- ALK is a receptor tyrosine kinase that is produced in a constitutively active form as a result of a gene rearrangement.
- In a subset of lung adenocarcinomas, a deletion on chromosome 5 fuses part of the ALK gene with EML4. The resulting fusion gene encodes a chimeric EML4-ALK protein with constitutive tyrosine kinase activity.

12. METASTATIC DEPOSITS

- Circulating tumor cells arrest and eventually form clinically significant metastatic deposits.
- This formation is related to three factors:
    - Location and vascular drainage of the primary tumor
    - Tropism of particular kinds of tumor cells for specific tissues
    - Escape from tumor dormancy.

13. SEED-SOIL HYPOTHESIS

- It was originally proposed by Paget
- Some tissues may provide a favourable “soil” for the growth of tumour seedlings.
- According to this “seed-soil” hypothesis, the ability of tumour cells originating from a particular site to adapt to a foreign environment may be limited to certain tissue types.

14. ENZYME ACTIVITY

- 09th edition - Benign tumors of the breast, colon, and stomach have little Type-IV collagenase activity, whereas their malignant counterparts overexpress this enzyme
- 10th edition - Benign tumors of the breast, colon, and stomach have little MMP-9 activity, whereas their malignant counterparts overexpress this enzyme.

15. THIRD STEP IN INVASION OF ECM

- 9th edition – Attachment to novel ECM components
- 10th edition – Attachment to remodeled ECM components

16. ONE-LINER:

- The sources of angiogenic factors include tumour cells, infiltrating inflammatory cells, or other tumour-associated stromal cells, and the ECM.
- Pediatric cancers are more likely to be caused by inherited mutations (particularly in tumour suppressor genes).
- Mitosis reflects the high rate of proliferation of neoplasm.
- Tumour suppressive genes are recessive traits
- Collection of cells and stroma composing new growths are referred to as neoplasms.
- Mutations in many genes contribute to tumorigenesis by interfering with host immune responses or altering interactions with the stroma, or by other mechanisms. By convention, these are not classified under driver and passenger mutations, since the terms are largely restricted to genes that influence the behaviour of the cells in a cell-intrinsic manner.
- In tumours, there is presence of cells with stem cell-like properties, which contribute to the relentless growth of metastatic lesions and to the “plasticity” that is needed for metastatic cells to adapt to growth in a new microenvironment.
- Tumour cells express anionic substances such as polyphosphate that activate factor XII (contact factor) which results in fibrin deposition and further stabilization of tumour emboli, which may enhance the ability of the cells to arrest en masse within capillary beds.
- Osteomalacia is a paraneoplastic syndrome due to phosphaturic mesenchymal tumour. The gene mutation responsible is FGF-23 gene.
- Protein antigens that elicit CD8+ cytotoxic T-cell responses are most relevant for protective antitumor immunity

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